Force, commitment, and perspective from Mandarin Chinese[*]

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Abstract

This paper offers a comprehensive bird's-eye view of the syntactic and semantic statuses of Mandarin question-forming strategies. Based on question markers' embeddability profiles in a series of nonquotational embedding contexts, we argue in favor of Dayal's (2025) three-way split in the left periphery, attributing the non-co-occurrence of question markers in different projections to their semantics. Specifically, we propose that (i) ForceP marks the starting point of clause-typing and houses the most embeddable *wh*-operator, *meiyou*, and A-not-A; (ii) PerspP triggers perspectival centering and houses question markers that are restrictedly embeddable (i.e., quasi-subordinated) like *bu*, *ma*, and *ne*; and (iii) CommitP modifies the speaker's commitment and houses the unembeddable *ba*. We advocate for a theory of selection that manifests a conspiracy of syntax and semantics by explicating that the embedding of PerspP-level question markers is restricted by two constraints: first, the embedded clause size must satisfy the syntactic selectional requirement of the embedding context; and second, the overall interpretation must be compatible with a not-at-issue requirement on the perspectival center's ignorance. Moreover, we fully flesh out the semantic account for the non-co-occurrence of question markers and argue for two fundamental distinctions in their semantic representations: one between assertions and questions and the other between monopolar and bipolar questions.

Keywords question marker • sentence-final particle • left periphery • illocutionary force phrase • perspective phrase • monopolar question • Mandarin Chinese

[*] We owe our deepest gratitude to Veneeta Dayal for her advice. We are also grateful to Jinlin Chen, Shifan Chen, Yulin Du, Lingxiao Gao, Xi Huang, Zhixian Huang, Manxin Lan, Yiming Li, Yunhao Li, Chengwei Liang, Nianpo Su, Ziqin Wei, Zhenrui Zhang, Yi Zhou, Zhenghao Zhou, Jiamu Zhu, Xiaomeng Zhu, and twelve anonymous survey participants for Mandarin data and judgments. For confirming some of the data and judgments in English, our thanks go to Richard Luo and Squid Tamar-Mattis. For invaluable discussion and feedback, we thank Éno Agolli, Comfort Ahenkorah, Athulya Aravind, Ushasi Banerjee, Christian Barker, Rajesh Bhatt, Carla Bombi, Claire Bowern, Mirium Butt, Simon Charlow, Qiushi Chen, Jarry Chia-Wei Chuang, Kathryn Davidson, Yoshiki Fujiwara, Daniel Hole, Laurence Horn, Miao-Ling Hsieh, Xuhui Hu, Kyle Johnson, Aidan Katson, Magdalena Kaufmann, Young-Hoon Kim, Si Kai Lee, Satoru Ozaki, María Mercedes Piñango, Paul Portner, Maribel Romero, Uli Sauerland, Zoltán Szabó, Milena Šereikaitė, Jason Shaw, Sze-Wing Tang, Alexander Wimmer, Jim Wood, Danfeng Wu, Qi Wu, Yimei Xiang, Yūsuke Yagi, Ka-Fai Yip, Xuetong Yuan, Florence Zhang-Yukun, Yuhan Zhang, four anonymous GLOW reviewers, and our audience at UConn Linguistics Graduate Roundtable 14, Yale Semantics Reading Group, TEAL 14, and GLOW 47. All errors are our own.

Contents

1 Introduction

Interrogative clauses vary in their ability to be embedded. In English, for instance, while *wh*-questions without subject-auxiliary inversion and polar questions introduced by *if* or *whether* are generally embeddable as in (1), questions in the form of a declarative with rising intonation (i.e., *rising declaratives*) are generally unembeddable as in (2). Below, we use "↑↑" to represent the rising intonation typically associated with English rising declaratives.

(1) a. I wonder/know/have forgotten [what we should do].

b. I wonder/know/have forgotten [whether it was raining].

(2) *I wonder/know/have forgotten [it was raining]$_{\uparrow\uparrow}$.

Compelling empirical observations and theoretical insights emerge when the variation in the embeddability of interrogative clauses is revealed to be more than a mere yes-or-no dichotomy. Recent work documents and theorizes a third possibility of question embeddability (Woods 2016, Dayal 2025) beyond the standard embeddable (nonroot) vs. unembeddable (root) distinction (a.o.: Emonds 1970, 1988, 2004, Hooper & Thompson 1973). In some English dialects, questions involving subject-auxiliary inversion are embeddable but constrained by independent conditions (Burling 1973, Henry 1995, McCloskey 2006) as in (3). These embedded constructions are known as *embedded inverted questions* (EIQs).

(3) a. I wonder [what should we do]. (McCloskey 2006: 89)

b. #I know [what should we do].

c. I have forgotten [what should we do].

Drawing on crosslinguistic evidence, Dayal (2025) accounts for the presence of the three possibilities (including EIQs) by positing a "CP < Persp(ective)P < S(peech)A(ct)P" hierarchy in the left periphery, a three-way distinction reminiscent of Coniglio & Zegrean's (2012) "C(lause)T(ype)P < Ill(ocutionaryForce)P" and Woods's (2016) "ForceP < I(llocutionary)A(ct)P." In Dayal's analysis, the bracketed questions in (1), (2), and (3) are of CP, SAP, and PerspP size, respectively. Essentially, in addition to a c-selectional constraint blocking the complementation of exceedingly large interrogative clauses like SAPs, which accounts for the ungrammaticality of (2), the proposal exploits a not-at-issue potential activeness (p-activeness) requirement introduced by PerspP, covering the pattern in (3). Dayal's proposal crucially provides a novel formal encoding of the division of labor among the multiple possible functions carried by the left periphery, offering revealing insights into where in the internal structure of a question the interrogative flavor enters.

This paper develops a point that Dayal hints at but does not elaborate on: the three-way division of the left periphery is systematically attested in Mandarin as well. To illustrate the three-way division of the Mandarin left periphery, the particle that plays a central role is *ma*, which marks polar questions. The syntactic, semantic, and pragmatic properties of *ma* have been extensively studied, often in connection with other question markers. To name several early references, see Chao 1926, 1965, Lü 1980, Hu 1981a,b, Li & Thompson 1981, Zhu 1982, T.-C. Tang 1988, and Cheng 1991. A recent synthetic analysis is offered by Pan (2015, 2017, 2018, 2019, 2022), who argues that *ma* occupies a peripheral projection dubbed *illocutionary force phrase* (iForceP), which hosts two other question markers: *meiyou* and *ba*.[1] Like *ma*, *meiyou* marks polar questions, but it additionally conveys perfective aspect with or without the perfective suffix *-le*. *Ba*, on the other hand, marks confirmation questions,

[1] See Appendix A.1 for a review of the different uses of *ma* and *ba*.

often translated into English as tag questions. Alongside the declarative base in (4), examples of these three question markers are presented in (5–7). Glossing abbreviations in this paper follow the Leipzig Glossing Rules.

(4) Zuotian xia-le yu.
yesterday fall-PFV rain
'It rained yesterday.'

(5) Zuotian xia(-le) yu **meiyou**?
yesterday fall-PFV rain MEIYOU
'Did it rain yesterday?'

(6) Zuotian xia-le yu **ma**?
yesterday fall-PFV rain MA
'Did it rain yesterday?'

(7) Zuotian xia-le yu **ba**?
yesterday fall-PFV rain BA
'It rained yesterday, didn't it?'

Although previous analyses that group these question markers within a single syntactic projection seem intuitive in light of their complementary distribution (e.g., Hu 1981a, Zhu 1982, Paul 2014, 2015, Pan 2015, 2017, 2018, 2019, 2022, Pan & Paul 2016, Paul & Pan 2016; cf. B. Li 2006, S.-W. Tang 2010a,b, Kim 2019), we argue instead that their non-co-occurrence stems from a semantic clash rather than from a syntactic competition, and each of the question markers occupies a distinct projection: "ForceP (for *meiyou*) < PerspP (for *ma*) < Commit(ment)P (for *ba*)," reflecting Dayal's "CP < PerspP < SAP" division. Our primary evidence comes from their embeddability profiles, where the prior discussion has largely converged on the generalization that *meiyou* is consistently embeddable, while *ma* and *ba* are not.[2] Drawing on both published data and collected judgments from 12 survey participants under experimental conditions, further corroborated by 14 additional native speakers of Mandarin, we refine this picture by showing that *ma* is in fact embeddable, albeit under restricted conditions. This results in a trichotomy in embeddability profiles of Mandarin question markers as stated in (8).

(8) a. *Meiyou* can be embedded by question-embedding predicates.
b. *Ma* can be embedded by question-embedding predicates under restricted conditions.
c. *Ba* cannot be embedded by question-embedding predicates.

We extend this trichotomy to other Mandarin question markers insofar as they qualify as such, which we mean to include the *wh*-operator, A-not-A, *bu*, *ne*, and rising declaratives. With this inventory of question markers, previous research has tended to focus either on the embeddability of multiple question markers within a nonexhaustive range of embedding contexts or on the embeddability of a single question marker. In this regard, on top of bringing to light a previously unnoticed third possibility of question marker embeddability, this paper contributes empirically by establishing a novel data paradigm for the embeddability of Mandarin interrogative clauses and by offering what is, to our knowledge, the first comprehensive bird's-eye view of both the syntactic classifications and the semantic properties of Mandarin question-forming strategies.

Building on the empirically established embeddability trichotomy and on Dayal's proposal of a three-way division in the left periphery — particularly the middle projection identified as PerspP, which introduces the p-activeness requirement — we account for the trichotomy in embeddability profiles of question markers by positing a joint contribution of syntax and semantics to the selection of complements. Syntactically, c-selectional constraints determine possible embedded clause sizes (or types), whereas semantically, PerspP imposes the p-activeness requirement. Question-embedding predicates may select a ForceP or a PerspP, but they

[2] In this paper, *embedding* is used to cover both regular subordination and quasi subordination (in the sense of Dayal & Grimshaw 2009 and Dayal 2025), while excluding quotation.

cannot embed CommitP, the highest projection among the three. Among embeddable interrogative clauses, the p-activeness requirement on the PerspP-level question markers checks whether the interpretation of the sentence is compatible with a context in which the question is p-active for the perspectival center, thereby regulating its embeddability, a constraint absent for ForceP question markers like *meiyou*. In this way, our proposal captures the observed trichotomy and correctly predicts that the embeddability of *ma* depends on two conditions: (i) the embedding context must license the appropriate clause size (or type), and (ii) the p-activeness requirement must be met. We summarize the central claims of our proposal in (9).

(9) a. The iForceP has a more fine-grained internal structure: ForceP < PerspP < CommitP.
b. The non-co-occurrence of the question markers in different projections is due to a semantic clash.
c. The question markers in the PerspP introduce the p-activeness requirement.

Viewed as a whole, the proposal yields a methodological takeaway: embeddability presents itself as a significantly more reliable test for cartographic analysis than non-co-occurrence, which may in fact reflect semantic rather than syntactic statuses. It also provides a new perspective on empirical domains where embeddability and non-co-occurrence appear to offer conflicting evidence for cartographic analysis, suggesting an explanation in terms of a division of labor between syntax and semantics. In this sense, although the present study focuses on a single language, the methodological framework it develops is readily applicable to other languages, and its theoretical implications are broadly crosslinguistic.

Beyond the three-way division in the left periphery, we demonstrate that an assertion-question distinction and a monopolar-bipolar distinction (cf. Biezma & Rawlins 2012, Krifka 2015, 2022, Rudin 2022, Matthewson 2025) are observationally present in the language and implementationally necessary for the theory. Presenting results from documented assertiveness diagnostics, as well as contrasts in the compatibility of question markers with contexts controlled for alternative saliency and with the attitude particle, the bias marker, and 'right' as a response, we argue that Mandarin question markers differ in whether they yield a proposition, a singleton set of propositions, or a plural set of propositions. Our findings therefore provide novel typological evidence for the crosslinguistically attested singleton-nonsingleton distinction in question semantics, and this paper offers the first such evidence from a language that has sentence-final particles (SFPs) and makes systematic use of them in question formation.

The rest of this paper is structured as follows. Section 2 reviews the traditional view on the cartography and embeddability of Mandarin question markers. Section 3 refines previous generalizations and arrives at a trichotomy in the embeddability profiles of these question markers. Section 4 presents a full analysis of their embeddability profiles, explicating how Dayal's three-way split in the left periphery is systematically manifested in Mandarin and offering a semantic account for their non-co-occurrence. Section 5 fully fleshes out the semantic account by establishing the assertion-question distinction and the monopolar-bipolar distinction. Section 6 concludes.

2 Traditional view

The traditional view of the syntactic statuses of *meiyou*, *ma*, and *ba* has not entertained the possibility that they each belong to their own projection, and a key observation motivating this view is that they never co-occur with each other, regardless of order. Examples are provided in (10–12).

(10) *Non-co-occurrence of* meiyou *and* ma
Intended: ‘Did it rain yesterday?’

a. *Zuotian xia-le yu **meiyou** **ma**?
yesterday fall-PFV rain MEIYOU MA

b. *Zuotian xia-le yu **ma** **meiyou**?
yesterday fall-PFV rain MA MEIYOU

(11) *Non-co-occurrence of* meiyou *and* ba
Intended: ‘Did it rain yesterday?’/‘It rained yesterday, didn’t it?’

a. *Zuotian xia-le yu **meiyou** **ba**?
yesterday fall-PFV rain MEIYOU BA

b. *Zuotian xia-le yu **ba** **meiyou**?
yesterday fall-PFV rain BA MEIYOU

(12) *Non-co-occurrence of* ma *and* ba
Intended: ‘Did it rain yesterday?’/‘It rained yesterday, didn’t it?’

a. *Zuotian xia-le yu **ma** **ba**?
yesterday fall-PFV rain MA BA

b. *Zuotian xia-le yu **ba** **ma**?
yesterday fall-PFV rain BA MA

Owing to the observed non-co-occurrence of question markers, prior studies have generally treated them as belonging to the same category and realizing the same functional head.[3]

2.1 Cartography of the left periphery

The discussion on the cartography of SFPs dates back to Hu (1981a) and Zhu (1982), who propose a tripartite classification of SFPs. They further observe that particles from different categories may co-occur but only in a fixed linear order as in (13), where we underline the category into which the question markers under discussion are classified.

(13) *Hu 1981a and Zhu 1982*

a. structural/tense particles < consonantal particles < vocalic particles (Hu 1981a: 348)

b. tense particles < imperative/interrogative particles < attitude/emotion particles (Zhu 1982: 207–214, summarized)

In their generalizations, since both *ma* and *ba* have consonantal onsets and convey imperative or interrogative force, they are classified as consonantal particles or imperative/interrogative particles and thus occupy the same position in the linear sequence. *Meiyou*, which is typically not considered a particle potentially due to its lexical semantic interpretation (composed of the perfective negation morpheme *mei* and the perfective marker *you*, as discussed in Cheng, Huang & Tang 1996, 1997, for example), is not included in their discussion.

The first fully articulated cartography of the Mandarin left periphery, as far as we are aware, is presented by B. Li (2006), and we reproduce her proposed hierarchy in (14).

[3] Note that (10a) should not be confused with a segmentally identical string where *ma* is replaced with ma_2 (discussed in Appendix A.1), which is acceptable under a different interpretation:

(i) Zuotian xia-le yu **meiyou** $\mathbf{ma_2}$?
yesterday fall-PFV rain MEIYOU MA_2
‘Just tell me, did it rain yesterday?’

(14) *B. Li 2006*
Fin(iteness)P < Foc(us)P < Deik(tisch)P < MoodP < Eval(uative)P < ForceP < DegreeP < Disc(ourse)P
(B. Li 2006: 178, adapted)

B. Li takes *ma* and *ba* to "indicate different degrees of the speaker's commitment" (B. Li 2006: 32) or of "the speaker's intention to have an action carried out" (B. Li 2006: 33). She treats both of them as degree markers and classifies them under DegreeP. With respect to *meiyou*, she argues against the Neg-to-C movement analysis proposed by Cheng, Huang & Tang (1996, 1997), contending instead that it is base-generated within the nonperipheral domain, and its sentence-final position results from deletion of its complement.

Before B. Li, the theoretical formulation of the Mandarin left periphery begins with S.-W. Tang (1998), who divides SFPs into *inner particles*, argued to realize T^0, and *outer particles*, argued to indicate the clause type following Cheng 1991. In S.-W. Tang 2010a,b, this division is further developed into a cartography comprising TP, CP, and three focus projections (FP1, FP2, and FP3) in order to capture the SFP ordering he identifies in (15).

(15) *S.-W. Tang 2010a,b*
tense particles < focus particles < degree particles < emotion particles (S.-W. Tang 2010b: 161)

S.-W. Tang observes that this ordering aligns with B. Li's cartography. Like B. Li, he treats *ma* and *ba* as degree particles and place them under FP2. The status of *meiyou* is not discussed.

In a parallel line of work, Paul (2005) demonstrates that the Mandarin left periphery aligns only partially with the hierarchy proposed by Rizzi (1997, 2004) and Belletti (2004). Inspired by Hu and Zhu, Paul (2014, 2015), Pan & Paul (2016), and Paul & Pan (2016) extend previous analyses of *ma* as a complementizer (e.g., H.-T. T. Lee 1986, Cheng 1991) and recast the tripartite classification and the fixed linear order as a split CP that has three layers as in (16).

(16) *Paul 2014, 2015, Pan & Paul 2016, and Paul & Pan 2016*
LowC(omplementizer)P < ForceP < Att(itude)P (Paul & Pan 2016: 56–68, summarized)

Under this proposal, both *ma* and *ba* function as "heads expressing the sentence type" (Paul 2014: 90) and are therefore analyzed as force complementizers, while *meiyou* is not discussed. In a similar vein, Erlewine (2017) proposes three classes of SFPs, with *ma* classified as a clause-typing SFP, positioned above low SFPs (argued to sit at the lower phase edge) and below attitude SFPs.

Building upon the work on the split Mandarin CP, Pan (2015, 2017, 2018, 2019, 2022) incorporates a substantially broader range of sentence-final elements and uses the co-occurrence ordering test to construct a more fine-grained representation of the Mandarin left periphery as in (17).

(17) *Pan 2015 et seq.*
S(entential)Asp(ect)P < OnlyP < i(llocutionary)ForceP < S(pecial)Q(uestion)P < Att(itude)P
(Pan 2022: 122, adapted)

Pan proposes a division of ForceP into iForceP and SQP. Since *meiyou*, *ma*, and *ba* are all "used to indicate the illocutionary force" (Pan 2015: 837), and they do not express "something else than a request for information" (Pan 2015: 843) in an interrogative form, Pan places them in the iForceP.

Importantly, no previous study has proposed splitting *meiyou*, *ma*, and *ba* into three distinct projections, a configuration that we ultimately argue for. Against this background, the next subsection reviews previous observations and analyses concerning the (un)embeddability of these question markers, recognizing it as the most crucial diagnostic for their cartographic status.

2.2 Dichotomy in question marker embeddability

Previous work (e.g., Li & Thompson 1981, Paul 2014, Pan 2015) observes that *meiyou* can always embed, while *ma* and *ba* never embed. This observation is illustrated in (18), where the question markers are embedded in the context of a noun complement or a clausal subject. Examples from the literature in which question markers are embedded within clausal objects are presented in (19).

(18) *Noun complement and clausal subject*
[Zuotian xia-le yu **meiyou/*ma/*ba**] (de wenti) hen guanjian.
Yesterday fall-PFV rain MEIYOU MA BA COMP question very crucial
(Intended:) '(The question of) whether it rained yesterday is very crucial.'

(19) *Clausal object*

a. Wo bu zhidao [ni chi-le fan **meiyou**].
1SG not know 2SG eat-PFV meal MEIYOU
'I don't know if you ate.' (Pan 2015: 841)

b. *Wo bu (xiang) wen ni [ta mingtian hui lai **ma**?]
1SG not want ask 2SG 3SG tomorrow will come MA
Intended: 'I don't (want to) ask you whether you will come.' (T.-C. Tang 1988: 297)

c. *Wo xiang zhidao [ni shi Xibanya ren **ba**].
1SG want know 2SG COP Spain person BA
Intended: 'I want to know whether you are Spanish.' (Pan 2015: 842)

The observed embeddability difference is accounted for by the lexical properties of question markers. For instance, Paul (2014) proposes that SFPs carry a [±root] feature, and only [−root] SFPs are embeddable. Under this proposal, *ma* and *ba* are [+root], and while Paul does not discuss *meiyou*, given its embeddability, it would be considered [−root]. The interpretability of the [±root] feature, however, is not addressed.

This lexical account would be called into question if *ma* can be shown to be embeddable after all, and establishing this is one of the empirical aims of this paper. We first draw attention to an example from the literature that has not received much discussion to the best of our knowledge. Consider (20).

(20) Wo wen ni [ta mingtian hui lai **ma**?]
1SG ask 2SG 3SG tomorrow will come MA
'I ask you whether he will come tomorrow.' (T.-C. Tang 1988: 295)

Although T.-C. Tang analyzes (20) as an instance of "direct discourse," which remains a possible interpretation of this specific example, we demonstrate in the next section that newly available data and diagnostics compel us to reconsider *ma* as indeed embeddable. This is where the third possibility of a question marker's embeddability becomes relevant, motivating a new analysis that supersedes the lexical account.

Before we move on, note that T.-C. Tang (1988) punctuates the sentence in (20) (and likewise back in (19b)) with a question mark, even though the main clause is declarative. This may reflect the fact that the subordinate clause is most naturally produced with the intonation pattern typical of Mandarin information-seeking *ma* questions. To avoid confusion, in what follows, we punctuate examples like (20) as in (21): the sentence ends with a full stop, but the interrogative intonation is indicated by an upward arrow.[4]

(21) Wo wen ni [ta mingtian hui lai **ma**]$_{\uparrow}$.
1SG ask 2SG 3SG tomorrow will come MA
'I ask you whether he will come tomorrow.' (T.-C. Tang 1988: 295, adapted)

3 New perspective

Another example that may challenge the lexical account's prediction of the embeddability of *ma* questions comes from Bhatt & Dayal (2020). We copy their example in (22).[5]

(22) John xiang zhidao [xia yu le **ma** zuotian]$_{\uparrow}$.
John want know fall rain PRF MA yesterday
'John wants to know whether it rained yesterday.' (Bhatt & Dayal 2020: 1122)

Contexts in which this example can be felicitously uttered include one in which the speaker acts as a proxy for the matrix subject, posing a question on their behalf and declaring their curiosity about the answer.

Prompted by Bhatt & Dayal, alongside the contexts in (23), we add in (24) examples that involve pronoun coreference and variable binding, which are evidence that the questions are not quotations and are truly embedded. If they were quotations, personal pronouns would behave differently (McCloskey 2006) as in (25), where the intended readings of the examples would not be compatible with the contexts in (23).

(23) a. *Context for (24a)*
(Xiaoming, Xiaohong, and you were classmates in college. At a class reunion after graduation, you and Xiaoming talked about Xiaohong, who was absent. Xiaohong had previously told you that she wanted to know who usually got higher grades between her and you during college. You also wanted to know, and you thought Xiaoming, who was the leader of the class, might know. Out of curiosity about the answer, you said to Xiaoming, …)

b. *Context for (24b)*
(Xiaoming, Xiaohong, Xiaogang, Xiaoqiang, and you were classmates in college. At a class reunion after graduation, you and Xiaoming talked about the other three, who were absent. Each of them had previously told you that she/he wanted to know who usually got higher grades between her/him

[4] We use "↑" in accordance with the convention in the literature, although we realize that the intonation associated with *ma* questions is not necessarily rising. See Liu & Xu 2005 for a detailed discussion of Mandarin question intonation. Note, however, that their terminology differs from ours: in particular, what they term *yes-no questions* correspond to what we call *rising declaratives*, while *ma* questions fall under their category of *particle questions*.

[5] Following Bhatt & Dayal's (2020) analysis, we bracket the example as if 'yesterday' forms part of the embedded question, but we acknowledge the possibility that the position of 'yesterday' may result from right dislocation (e.g., Chao 1965, Lu 1980, Cheung 2009, T. T.-M. Lee 2017, Yip 2025). It could, in fact, occupy a position in the matrix periphery. Nevertheless, neither their nor our argument depends on the precise analysis of the position of 'yesterday', as the sentence remains acceptable and retains the intended interpretation when 'yesterday' appears in its default position:

(i) John xiang zhidao [zuotian xia yu le **ma**]$_{\uparrow}$.
John want know yesterday fall rain PRF MA
'John wants to know whether it rained yesterday.'

and you during college. You also wanted to know, and you thought Xiaoming, who was the leader of the class, might know. Out of curiosity about the answer, you said to Xiaoming, …)

(24) *Embedded questions: 1SG = speaker and 3SG = matrix subject*

a. Xiaohong$_i$ xiang zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_\uparrow$.
Xiaohong want know 3SG POSS score than 1SG high MA
'Xiaohong$_i$ wants to know whether her$_i$ score is higher than mine$_j$.'

b. [Mei ge ren]$_i$ dou xiang zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_\uparrow$.
every CLF person all want know 3SG POSS score than 1SG high MA
'Everyone$_i$ wants to know whether their$_i$ score is higher than mine$_j$.'

(25) *Quoted questions: 1SG = quoting predicate subject and 3SG = someone else*

a. Xiaohong$_i$ xiang zhidao: ["ta$_j$ de fenshu bi wo$_i$ gao **ma**?"]
Xiaohong want know 3SG POSS score than 1SG high MA
'Xiaohong$_i$ wants to know, "is her$_j$ score higher than mine$_i$?"'

b. [Mei ge ren]$_i$ dou xiang zhidao: ["ta$_j$ de fenshu bi wo$_i$ gao **ma**?"]
every CLF person all want know 3SG POSS score than 1SG high MA
'Everyone$_i$ wants to know, "is their$_j$ score higher than mine$_i$?"'

Data resembling (24) have been confirmed in Beijing Mandarin under experimental settings by Y. Liu (2025) using a novel semantic judgment elicitation paradigm ($n = 12$) and further corroborated through personal communication with more Beijing Mandarin speakers ($n = 5$).[6] While we believe that the acceptability is found in a much wider range of dialects of Mandarin — including those spoken in Fuzhou, Hanzhong, Hong Kong, Mianyang, Nanjing, Nanning, Wenzhou, Wuhan, and Xi'an ($n = 9$), whose speakers we have consulted — we acknowledge regional variations, which we leave for future work.[7]

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[6] To briefly summarize the experimental setup, Y. Liu's (2025) novel semantic judgment elicitation paradigm presented participants with the following instruction:

(i) Below are some pairs of expressions, each with an accompanying context. In each pair of expressions, option (a) consists of two sentences, and option (b) attempts to convey the same meaning using only one sentence. Your task is to understand the accompanying context, read both expressions aloud, and decide whether the two sentences in (a) can be naturally conveyed using the single sentence in (b) in the accompanying context. If you believe it is possible, choose (b). If not, choose (a). If both (a) and (b) seem unacceptable for you, choose (a) as well. (Y. Liu 2025: 94)

For each item under each condition, the design essentially forced participants to choose between the target sentence and a two-sentence paraphrase of it, in which the first sentence conveyed the matrix content, and the second conveyed the embedded content. To illustrate, the two-sentence paraphrase of (24a) in the experiment is given below:

(ii) Xiaohong$_i$ xiang zhidao yi jian shi. Na jiu shi [ta$_i$ de fenshu shi-bu-shi bi wo$_j$ gao].
Xiaohong want know one CLF thing that just COP 3SG POSS score COP-not-COP than 1SG high
'Xiaohong$_i$ wants to know one thing. That thing is whether her$_i$ score is higher than mine$_j$.' (Y. Liu 2025: 94)

Subscripts here are only for clarity and were not included in the survey. Pronoun referents were made explicit in the accompanying context (for which (24a) serves as an example). The results revealed a significant contrast in the acceptability of embedding *ma* questions between quasi subordinators and non–quasi subordinators, as well as between matrix declaratives and matrix interrogatives. We discuss these contrasts in Section 3.1.

[7] As an illustration of regional variations, Taiwanese speakers may find embedded *ma* questions much less acceptable even in contexts where we predict them to be possible, as reported by Pan (2015, 2019) a.o.:

The judgments are additionally supported by naturally occurring examples in the BCC Corpus (Xun et al. 2016). In both examples cited in (26), the surrounding context makes it clear that the 3SG personal pronoun (or one of the 3SG personal pronouns) in the embedded question refers to the matrix subject. Although gender information is not explicitly marked in the transcription below, it is conveyed through the distinct characters used in the corpus. Bracketing, boldface, and subscripts are solely for the readers' convenience and are not present in the original data. The sentence-final question mark in (26a) is retained.

(26) a. *("Ni ai wo ma?" "Ni ne?" …)*
[("Do you love me?" "What about you?" …)]

Ta$_i$ ce shou fan-wen, ye hao xiang hao xiang zhidao [ta$_j$ ai ta$_i$ **ma**?]
3FSG turn head return-ask also really want really want know 3MSG love 3FSG MA
'She turned her head and asked in return, also really, really wanting to know whether he loved her.' (BCC Corpus)

b. Yaoshi [na ge niu]$_i$ zai wen wo [yuanyi deng ta$_i$ **ma**], wo lian sikao de
if which CLF girl again ask 1SG willing wait 3FSG MA 1SG even think LNK
jihui dou bu hui gei, …
chance all not will give
'If any girl asks me again whether [I]'m willing to wait for her, I won't even give [myself] a chance to think about it, …' (LNK = linker) (BCC Corpus)

3.1 Trichotomy in question marker embeddability

The embeddability of *ma* depends heavily on the matrix predicate type. While predicates like 'want to know' accept it, other predicates like 'know' reject it as in (27).

(27) #Xiaohong$_i$ zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_{(\uparrow)}$.
Xiaohong know 3SG POSS score than 1SG high MA
Intended: 'Xiaohong$_i$ knows whether her$_i$ score is higher than mine$_j$.'

Building on Woods (2016) and Dayal's (2025) observations of English EIQs, we find that the observed contrast between 'want to know' and 'know' does not align with the familiar rogative-responsive distinction. This can be seen from predicates like 'forget', which is responsive but patterns with 'want to know' instead of 'know', as it allows the embedding of *ma* (28).

(i) a. *Ta wen wo [ni xihuan Faguo ma].
3SG ask 1SG 2SG like France MA
Intended: 'He asks me whether you like France.' (Pan 2015: 840)

b. *Ta xiang zhidao [ni qu-guo Faguo ma].
3SG want know 2SG go-EXP France MA
Intended: 'He wants to know whether you have been to France.' (EXP = experiential) (Pan 2019: 100)

Both examples cited here are acceptable in Beijing Mandarin.

We have also consulted Mandarin speakers who are also native to the Jiaxing or Shanghainese dialect of Wu Chinese, and they find *ma* questions uniformly embeddable. We leave a more systematic investigation of these data points to future work.

(28) (Xiaoming, Xiaohong, and you were classmates in college. At a class reunion after graduation, you and Xiaoming talked about Xiaohong, who was absent. Xiaohong had previously told you she'd forgotten who usually got higher grades between you and her during college. You had also forgotten, but you thought Xiaoming, who was the leader of the class, might remember. Out of curiosity about the answer, you said to Xiaoming, …)

Xiaohong$_i$ wangji-le [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_\uparrow$.
Xiaohong forget-PFV 3SG POSS score than 1SG high MA
'Xiaohong$_i$ forgot whether her$_i$ score is higher than mine$_j$.'

The nature of this contrast has to do with whether the predicate is semantically compatible with a context in which the possibility that the matrix subject does not know the answer to the embedded question exists: *ma* necessitates the manifestation of this possibility in the context, a condition we refer to as *ignorance requirement* (IR). We follow Dayal & Grimshaw (2009) and term a predicate that satisfies the IR (e.g., 'want to know') a *quasi subordinator* (QS) and one that does not (e.g., 'know') a *non–quasi subordinator* (NQS). An illustration of the contrast between QS and NQS in terms of the IR as well as a comparison with rogative and responsive predicates is presented in Table 1.[8]

Table 1 Question embedding predicates of different types

	QS (✓IR): $\neg\text{know}(x, \text{answer}(Q))$	NQS (✗IR): $\Box\text{know}(x, \text{answer}(Q))$
Rogative: [+Q]	*ask*, *want to know*, *wonder*	
Responsive: [±Q]	*forget*	*explain*, *know*, *remember*

Additional support for the IR arises from the way an NQS exhibits shiftiness under matrix modality (Danfeng Wu, pers. comm.), negation, and interrogative. Although the degree of acceptability shows minor variation across speakers for some of the examples, all consultants agree that (29–31) present a clear contrast with (27) in appropriate contexts.[9]

(29) *Modality*

a. (?)Huoxu/yexu Xiaohong$_i$ zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_\uparrow$.
maybe maybe Xiaohong know 3SG POSS score than 1SG high MA
'Maybe Xiaohong$_i$ knows whether her$_i$ score is higher than mine$_j$.'

b. (?)Xiaohong$_i$ keneng zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_\uparrow$.
Xiaohong may know 3SG POSS score than 1SG high MA
'Xiaohong$_i$ may know whether her$_i$ score is higher than mine$_j$.'

[8] For simplicity, we treat 'want to know' as a single unit here, but Section 4.2 will discuss 'want to know' in compositional terms.

[9] In (31), the matrix interrogative makes use of another strategy for forming polar questions in Mandarin — namely, the A-not-A template. Section 3.2 discusses its embeddability profile in detail.

(30) *Negation*
Xiaohong$_i$ bu zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_\uparrow$.
Xiaohong not know 3SG POSS score than 1SG high MA
'Xiaohong$_i$ doesn't know whether her$_i$ score is higher than mine$_j$.'

(31) *Interrogative*
Xiaohong$_i$ zhi-bu-zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_\uparrow$?
Xiaohong know-not-know 3SG POSS score than 1SG high MA
'Does Xiaohong$_i$ know whether her$_i$ score is higher than mine$_j$?'

Even though the matrix predicate is an NQS and cannot satisfy the IR by itself, the presence of matrix modality, negation, or interrogative opens up the possibility that the matrix subject is ignorant of the answer to the embedded question, thereby satisfying the IR. Among these conditions, the effect of the matrix interrogative has been confirmed under experimental settings by Y. Liu (2025). See McCloskey 2006, Woods 2016, and Dayal 2025 for a more detailed discussion of these conditions in connection with English EIQs.

The embeddability profile we identify for *ma* appears to be new to the Chinese linguistics literature, especially in light of its semantic nature. This motivates a reassessment of the embeddability of other question markers, including *meiyou* and *ba*. Unlike *ma*, their (un)embeddability remains unaffected regardless of whether the IR is satisfied and aligns with the traditional view: *meiyou* questions (32) are always embeddable, while *ba* questions (33) never are (although *ba* can occur in quotations).[10]

(32) a. Xiaohong$_i$ xiang zhidao [ta$_i$ jige-le **meiyou**]. *QS* + meiyou *question*
Xiaohong want know 3SG pass-PFV MEIYOU
'Xiaohong wants to know if he passed.'

b. Xiaohong$_i$ zhidao [ta$_i$ jige-le **meiyou**]. *NQS* + meiyou *question*
Xiaohong know 3SG pass-PFV MEIYOU
'Xiaohong knows if he passed.'

[10] The perfective aspect of *meiyou* makes 'whether his score is higher than mine' sound unnatural with it. While (32), as a result, fails to maintain a minimal pair relation with the *ma* question examples presented thus far, the embeddability profile of *ma* questions can be replicated using 'whether he passed':

(i) a. Xiaohong$_i$ xiang zhidao [ta$_i$ jige-le **ma**]$_\uparrow$. *QS* + ma *question*
Xiaohong want know 3SG pass-PFV MA
'Xiaohong wants to know whether he passed.'

b. #Xiaohong$_i$ zhidao [ta$_i$ jige-le **ma**]$_\uparrow$. *#NQS* + ma *question*
Xiaohong know 3SG pass-PFV MA
Intended: 'Xiaohong knows whether he passed.'

In the remainder of the paper, particularly when we turn to question markers beyond *meiyou*, *ma*, and *ba*, it should be noted that some other question markers may also exhibit some degree of idiosyncrasy as *meiyou* does here. Consequently, no single declarative base can serve uniformly across all question markers. Nonetheless, we have endeavored to keep the relevant examples as minimally different from each other as possible.

(33) a. *Xiaohong$_i$ xiang zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ba**]. **QS* + ba *question*
Xiaohong want know 3SG POSS score than 1SG high BA
Intended: 'Xiaohong$_i$ wants to know whether her$_i$ score is higher than mine$_j$.'

b. *Xiaohong$_i$ zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ba**]. **NQS* + ba *question*
Xiaohong know 3SG POSS score than 1SG high BA
Intended: 'Xiaohong$_i$ knows whether her$_i$ score is higher than mine$_j$.'

We thus observe a trichotomy in the (un)embeddability of question markers in clausal objects. Together with the previous observations concerning noun complements and clausal subjects, we summarize the embeddability profiles of *meiyou*, *ma*, and *ba* in Table 2.[11]

Table 2 Embeddability profiles of *meiyou*, *ma*, and *ba*

	Meiyou	*Ma*	*Ba*
Noun complement	✓	✗	✗
Clausal subject	✓	✗	✗
Clausal object (IR not satisfied)	✓	✗	✗
Clausal object (IR satisfied)	✓	✓	✗

In what follows, we extend the observed trichotomy to other Mandarin question-forming strategies. We examine the embeddability profiles of other question markers identified by Paul 2014 et seq. in ForceP and by Pan 2015 et seq. in iForceP, including the *wh*-operator, *bu*, and *ne*. Another commonly used polar question-forming strategy, A-not-A, would also merit a reassessment of its embeddability. We then conclude the section with a discussion of rising declaratives.

3.2 Question markers with *meiyou*-type embeddability

Mandarin *wh*-questions are embeddable in all question embedding contexts (e.g., Huang 1982a,b, Cheng 1991) as in (34). They pattern like *meiyou* in terms of embeddability.

(34) a. *Noun complement or clausal subject*
[Mingtian **shenme shihou** xia yu] (de wenti) hen guanjian.
tomorrow what time fall rain COMP question very crucial
'(The question of) when it will rain tomorrow is very crucial.'

b. *Clausal object*
Xiaohong$_i$ (xiang) zhidao [ta$_i$ de fenshu **weishenme** bi wo$_j$ gao].
Xiaohong want know 3SG POSS score why than 1SG high
'Xiaohong$_i$ knows/wants to know why her$_i$ score is higher than mine$_j$.'

[11] The DP shell analysis (Lees 1960, Rosenbaum 1965, Davies & Dubinsky 1998, Takahashi 2010, Hartman 2012) claims that clausal subjects are DPs, formed by phonologically null D heads that embed CPs. Under this analysis, clausal subjects may be understood to form a subset of noun complements. Our proposal does not yield any predictions that conflict with the DP shell analysis, as we find no instances in which a question marker's embeddability differs between noun complements and clausal subjects.

A-not-A signals polar questions. It is formed by reduplicating the verb or its leftmost portion and placing a negative morpheme in between (e.g., Li & Thompson 1979, Huang 1982b, 1991, Zhu 1985) as in (35).[12]

(35) Mingtian **xia-bu-xia** yu?
tomorrow fall-not-fall rain
'Will it rain tomorrow?'

As shown in (36), A-not-A questions are also embeddable across the board, resembling *meiyou* questions.

(36) a. *Noun complement or clausal subject*
[Mingtian **xia-bu-xia** yu] (de wenti) hen guanjian.
tomorrow fall-not-fall rain COMP question very crucial
'(The question of) whether it will rain tomorrow is very crucial.'

b. *Clausal object*
Xiaohong$_i$ (xiang) zhidao [ta$_i$ **ji-mei-jige**].
Xiaohong want know he pass-not.PFV-pass
'Xiaohong$_i$ knows/wants to know whether she$_i$ passed.'

Following previous analyses, we assume that both *wh-* (e.g., Y.-H. A. Li 1992, Aoun & Li 1993a,b, Tsai 1994, Soh 2005) and A-not-A (e.g., Ernst 1994, Schaffar & Chen 2001, Law 2006, Tsai & Yang to appear) questions involve a dependency with a functional head or an operator bearing some interrogative clause-typing feature (e.g., [+Q]) in the left periphery. Any subsequent claims regarding the cartography of *wh-* or A-not-A questions are directed toward this left peripheral element that bears the clause-typing feature.

3.3 Question markers with *ma*-type embeddability

The negation morpheme *bu* 'not' can be used as an SFP and signal polar questions (Cheng, Huang & Tang 1996, 1997, Pan 2019, Chen & Huang 2020) as in (37).

(37) Mingtian xia yu **bu**?
tomorrow fall rain BU
'Will it rain tomorrow?'

We observe that *bu* questions exhibit the same embeddability pattern as *ma* questions, as presented in (38). They do not embed under noun complements or clausal subjects, but they may embed under clausal objects only when the IR is satisfied.

(38) a. *Noun complement or clausal subject*
*[Mingtian xia yu **bu**]$_{(\uparrow)}$ (de wenti) hen guanjian.
tomorrow fall rain BU COMP question very crucial
Intended: '(The question of) whether it will rain tomorrow is very crucial.'

[12] The literature further distinguishes variants of A-not-A where the reduplicated morpheme is a modal (e.g., *hui* 'will') or a focus marker (lexically identical to the copula *shi*), often referred to as *M-not-M* and *B-not-B* (e.g., Hagstrom 2006), respectively. B-not-B is further set apart from the others as *outer A-not-A* in contrast to *inner A-not-A* (e.g., Tsai & Yang 2015, to appear). These variants do not differ in their embeddability profiles, and we discuss A-not-A in its strictest sense, excluding constructions that superficially resemble A-not-A but have been argued not to involve reduplication (e.g., VP-not-V) and referring only to the inner type.

b. *Clausal object (IR not satisfied)*
#Xiaohong$_i$ zhidao [ta$_i$ de fenshu bi wo$_j$ gao **bu**]$_{(\uparrow)}$.
Xiaohong know 3SG POSS score than 1SG high BU
Intended: 'Xiaohong$_i$ knows whether her$_i$ score is higher than mine$_j$.'

c. *Clausal object (IR satisfied)*
Xiaohong$_i$ xiang zhidao [ta$_i$ de fenshu bi wo$_j$ gao **bu**]$_\uparrow$.
Xiaohong want know 3SG POSS score than 1SG high BU
'Xiaohong$_i$ wants to know whether her$_i$ score is higher than mine$_j$.'

Ne can be attached to interrogatives such as *wh*-, A-not-A, and *meiyou* questions, but not to *bu*, *ma*, or *ba* questions.[13] In general, it signals a "follow-up question" (Paul 2014: 83) or "marks that the current discourse move is 'unexpected'" (X. Yuan 2023: 748; cf. Chu 2009). When the interrogative base is a polar question, *ne* can give "a meaning along the lines of an English *or not* question" (Constant 2014: 339; cf. Dong 2018). An example of its usage with *meiyou* is in (39).

(39) Zuotian xia-le yu meiyou **ne**?
yesterday fall-PFV rain MEIYOU NE
'Did it rain yesterday or not?'

The embeddability profile of *ne* resembles that of *bu* and *ma*. It is embeddable in a clausal object when the IR is satisfied but unembeddable elsewhere as in (40).

(40) a. *Noun complement or clausal subject*
*[Zuotian xia-le yu meiyou **ne**]$_{(\uparrow)}$ (de wenti) hen guanjian.
yesterday fall-PFV rain MEIYOU NE COMP question very crucial
Intended: '(The question of) whether it rained yesterday or not is very crucial.'

b. *Clausal object (IR not satisfied)*
#Xiaohong$_i$ zhidao [ta$_i$ jige-le meiyou **ne**]$_{(\uparrow)}$.
Xiaohong know 3SG pass-PFV MEIYOU NE
Intended: 'Xiaohong$_i$ knows whether she$_i$ passed or not.'

c. *Clausal object (IR satisfied)*
Xiaohong$_i$ xiang zhidao [ta$_i$ jige-le meiyou **ne**]$_\uparrow$.
Xiaohong want know 3SG pass-PFV MEIYOU NE
'Xiaohong$_i$ wants to know whether she$_i$ passed or not.'

3.4 Questions that do not embed

We adopt the arguments advanced by Pan & Paul (2016) and Luo (to appear) and maintain that none of the question markers discussed thus far are "optional" in the sense of Cheng 1991, Bailey 2010, 2013, and Biberauer, Holmberg & Roberts (2014). Based on this commitment, rising declaratives (RDs) in Mandarin should be analyzed on a par with other question markers rather than as an intonational pattern that invariably accompanies interrogatives. Like in (2) for English, we use a double upward arrow to represent the intonation typically associated with RDs in Mandarin as well, as presented in (41).

[13] See Appendix A.2 for a review of the different uses of *ne*.

(41) [Mingtian xia yu?]$_{\uparrow\uparrow}$
tomorrow fall rain
'It will rain tomorrow?'

As in (42), similar to *ba* questions, rising declaratives do not embed.[14]

(42) a. *Noun complement or clausal subject*
*[Mingtian xia yu]$_{\uparrow\uparrow}$ (de wenti) hen guanjian.
tomorrow fall rain COMP question very crucial
Intended: '(The question of) whether it will rain tomorrow is very crucial.'

b. *Clausal object*
*Xiaohong$_i$ (xiang) zhidao [ta$_i$ de fenshu bi wo$_j$ gao]$_{\uparrow\uparrow}$.
Xiaohong want know 3SG POSS score than 1SG high
Intended: 'Xiaohong$_i$ knows/wants to know whether her$_i$ score is higher than mine$_j$.'

4 Conspiracy of syntax and semantics

The embeddability profiles covered thus far are summarized in Table 3. The observed trichotomy, at the very least, poses a serious empirical challenge to lexical analyses based on a binary feature such as [±root], which by itself can account only for a two-way distinction. Alternative analyses that posit multiple features are also undermined by the semantic dimension inherent in the embeddability profile of *bu*, *ma*, and *ne*, not to mention the potential complications associated with the interpretability of the features. These considerations motivate a new analysis, one that integrates both syntactic and semantic components.

Table 3 Embeddability profiles of question markers

	Wh-	*Meiyou*	A-not-A	*Bu*	*Ma*	*Ne*	*Ba*	↑↑
Noun complement	✓	✓	✓	✗	✗	✗	✗	✗
Clausal subject	✓	✓	✓	✗	✗	✗	✗	✗
Clausal object (IR not satisfied)	✓	✓	✓	✗	✗	✗	✗	✗
Clausal object (IR satisfied)	✓	✓	✓	✓	✓	✓	✗	✗

[14] Another type of questions that do not embed in Mandarin is those followed by *ma*$_2$ (introduced in Appendix A.1). The relevant data are presented below:

(i) a. *Noun complement or clausal subject*
*[Zuotian xia-le yu meiyou **ma$_2$**] (de wenti) hen guanjian.
yesterday fall-PFV rain MEIYOU MA$_2$ COMP question very crucial
Intended: '(The question of) whether it rained yesterday is very crucial.'

b. *Clausal object*
*Xiaohong$_i$ (xiang) zhidao [ta$_i$ jige-le meiyou **ma$_2$**].
Xiaohong want know 3SG pass-PFV MEIYOU MA$_2$
Intended: 'Xiaohong$_i$ knows/wants to know whether she$_i$ passed.'

The fact that *ma* and *ma*$_2$ differ in their embeddability supports the view that they are not a single lexical item, aligning with Pan's (2019) classification of these particles as belonging to distinct projections (i.e., *ma* in iForceP and *ma*$_2$ in AttitudeP).

4.1 Syntactic component: Split iForceP

We capture the tiered embeddability pattern using a three-way split iForceP and label the projections as force phrase (ForceP), perspective phrase (PerspP), and commitment phrase (CommitP). With respect to the grammatical functions and the nomenclature of the projections, we adopt ForceP in the sense of Rizzi 1997, 2001, 2004 (and Paul 2014, 2015 to some extent), PerspP in the sense of Dayal 2025, and CommitP in the sense of Miyagawa 2022 and Miyagawa & Hill 2025. As will become clearer in the remainder of this paper, ForceP marks the earliest stage of clause typing, PerspP triggers perspectival centering, and CommitP accommodates elements that modify the speaker's commitment to the core proposition. In this respect, ForceP maps roughly onto Dayal's CP, PerspP maps more closely onto Dayal's PerspP, and CommitP, together with any projections above it, maps roughly onto Dayal's whole SAP domain. The periphery that we propose looks like (43).[15]

(43) ForceP < PerspP < CommitP

a. ForceP: *wh*-, *meiyou*, and A-not-A

b. PerspP: *bu*, *ma*, and *ne*

c. CommitP: *ba* and ↑↑

A neat correspondence emerges between the grammatical function of each of the three projections and its degree of embeddability. ForceP hosts the most embeddable elements, including the *wh*-operator, A-not-A, and *meiyou*. PerspP hosts elements embeddable under restricted conditions, including *bu*, *ma*, and *ne*. CommitP hosts the unembeddable element *ba*, and we contend that the realization of a rising declarative requires the size of CommitP.

Table 4 Co-occurrence of question markers

		ForceP			PerspP			CommitP
		Wh-	*Meiyou*	A-not-A	*Bu*	*Ma*	*Ne*	*Ba*
CommitP	*Ba*	✗	✗	✗	✗	✗	✗	
	Ne	✓	✓	✓	✗	✗		
PerspP	*Ma*	✗	✗	✗	✗			
	Bu	✗	✗	✗				
	A-not-A	✗	✗					
ForceP	*Meiyou*	✗						
	Wh-							

[15] Our Mandarin left periphery is compatible with the hierarchy proposed in Paul 2014 et seq. and Pan 2015 et seq. and can be incorporated into Pan's cartography as shown below:

(i) SAspP < OnlyP < ForceP < PerspP < CommitP < SQP < AttitudeP

It is incompatible with the hierarchy proposed by B. Li (2006) and S.-W. Tang (2010a,b), which assumes a unified treatment of *ma* and ma_2 as well as of *ba* and $ba_{2\text{–}4}$ (introduced in Appendix A.1).

Recall that most of the aforementioned question markers cannot co-occur with each other, as presented in Table 4.[16] Since our split iForceP proposal eliminates the syntactic competition between *bu* & *ma* and question markers in ForceP, as well as between *ba* and all other question markers, it calls for a new analysis of their complementary distribution. We attribute their non-co-occurrence to their semantics: the output of a lower question marker is not of the right type to serve as an input of any higher question markers. Deferring a deeper discussion to later sections, we briefly note here that we adopt a Karttunen (1977) analysis for question semantics: we assume that a proposition is of type st, while a question is of type $\langle st, t\rangle$ (i.e., a set of propositions). Moreover, we propose that the *wh*-operator, A-not-A, *meiyou*, *bu*, and *ma* take propositions and yield questions, while *ne* takes and yields questions. As for *ba*, we propose that it takes and yields propositions based on its assertoric nature argued by X. Yuan (2021) and Ye (2024), a point we elaborate on in Section 5.2. We summarize the semantic types of the input and output of question markers in Table 5.

Table 5 Semantic types of the input and output of question markers

ForceP	PerspP	CommitP
$st \rightarrow$ *wh-* $\rightarrow \langle st, t\rangle$	$st \rightarrow$ *bu* $\rightarrow \langle st, t\rangle$	$st \rightarrow$ *ba* $\rightarrow st$
$st \rightarrow$ *meiyou* $\rightarrow \langle st, t\rangle$	$st \rightarrow$ *ma* $\rightarrow \langle st, t\rangle$	
$st \rightarrow$ A-not-A $\rightarrow \langle st, t\rangle$	$\langle st, t\rangle \rightarrow$ *ne* $\rightarrow \langle st, t\rangle$	

Since all question markers except for *ne* take propositions, while all question markers except for *ba* yield sets of propositions, they cannot co-occur with each other due to type clash. With respect to *ba*, although it yields propositions, which are type-wise acceptable inputs to other question markers, it belongs to the highest one among the three projections and cannot feed its output to a lower question marker. The upshot is that the non-co-occurrence of question markers can be accounted for under our proposal as long as we assume a distinction in semantic representation between propositions and questions.

4.2 Semantic component: P-activeness

The second component of our analysis argues that PerspP should be interpreted à la Dayal 2025 in the sense that it triggers perspectival centering and introduces a requirement on the embedded question's potential activeness for the perspectival center. In Dayal's (2025) proposal, the perspective head in interrogatives poses a not-at-issue requirement that the complement question must be potentially active (p-active) for the perspectival center. Two concepts merit further clarification: *perspectival center* and *p-activeness*.

Roughly speaking, the perspectival center is "the one who holds the question." In Dayal's implementation, the perspectival center is syntactically introduced in the specifier position of PerspP by a phonologically null anaphor (i.e., PRO) that must be bound. In a matrix question, this anaphor is bound by the speaker. When the PerspP is embedded as in (44), the anaphor is bound by a matrix argument, typically the matrix subject.

[16] Without the aid of phonetic experiments, it is impossible to determine whether the rising intonation in rising declaratives co-occurs with other question markers. We therefore leave its (non-)co-occurrence pattern for future work.

(44) *Embedded PerspP: Perspectival center = matrix argument*
(The question of whether it rained is p-active for the matrix subject John.)

John$_i$ xiang zhidao [$_{\text{PerspP}}$ PRO$_i$ [$_{\text{Persp}'}$ xia yu le [$_{\text{Persp}^0}$ **ma**]]]$_\uparrow$.
John want know fall rain PRF MA
'John wants to know whether it rained.' (Bhatt & Dayal 2020: 1122, adapted)

For a question Q to be p-active for the perspectival center x, a minimal requirement is the possibility of x's ignorance about the answer to Q as in (45), or the IR as discussed in Section 3.1.[17] This not-at-issue requirement is formalized as a lexical property of $\text{Persp}^0_{\text{CQ}}$, as copied in (46), where CQ stands for *centered question*.[18]

(45) *Minimal requirement for p-activeness*
$\forall Q\,[\forall x\,[\text{p-active-for}(Q, x) \rightarrow \neg\text{know}(x, \text{answer}(Q))]]$ (Dayal 2025: 673)

(46) *Denotation of* Persp^0_{CQ}
$[\![\text{Persp}^0_{\text{CQ}}]\!] = [\lambda Q.\, \lambda x\!: \text{p-active-for}(Q, x).\, Q]$ (Dayal 2025: 673)

A key point of divergence between our framework and Dayal's lies in the input type of $\text{Persp}^0_{\text{CQ}}$. In Dayal's system, $\text{Persp}^0_{\text{CQ}}$ in interrogatives takes questions as its input, as interrogative clause-typing uniformly takes place at CP (roughly corresponding to our ForceP). In contrast, for Mandarin PerspP question markers, we slightly modify the denotation to allow certain markers to take propositions as input. This is illustrated in (47) using *ma* as an example, where we use $\text{PR}(x)$ to abbreviate the p-activeness requirement for x.[19]

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[17] Apart from the IR as a minimal requirement, possible candidates for the PR's pragmatic component include *resolution requirement* (i.e., "whether a positive answer to the matrix question in each case can lead to a resolution of the question that the speaker is interested in" (Dayal 2025: 685)), *knowledge requirement* (i.e., whether "[i]t is possible that the addressee knows the answer' (Y. Liu 2025: 93)), or the perspectival center's curiosity toward the question, which may help account for the unacceptability of examples such as (19b).

[18] Dayal (2025) suggests that the three-layered peripheral structure is present in declaratives and imperatives as well, where the perspective head is realized as $\text{Persp}^0_{\text{CP}}$ (*centered proposition*) and $\text{Persp}^0_{\text{CI}}$ (*centered imperative*), respectively. We adopt these variants of the perspective head at face value and refer interested readers to Dayal 2025.

[19] The denotation we propose for *ma* is consistent with the observation that (i) question markers below *ma* cannot co-occur with it (cf. Sections 2 and 4.1), and (ii) *wh*-expressions under the scope of *ma* cannot be interpreted as questions:

(i) a. *Zuotian **xia-mei-xia** yu **ma**? *A-not-A* + ma ⇝ ✗
yesterday fall-not.PFV-fall rain MA
Intended: 'Did it rain yesterday?'

b. Zuotian **shenme shihou** xia-le yu **ma**? wh + ma ⇝ ✓*indefinite*, ✗*question*
yesterday what time fall-PFV rain MA
'Did it rain sometime yesterday?'
Not: 'When did it rain yesterday?'

An alternative analysis where *bu* and *ma* are identity functions that select a phonologically null clause-typing head would require extra assumptions to explain their nonoptionality (in the sense of Section 3.4). One could also posit that *bu* and *ma* are force heads that obligatorily project to PerspP, or that they expone a fused force-perspective head. Without further empirical motivation, we set these possibilities aside.

Also in (47), we follow Dayal's (2025) treatment of the PR as a presupposition, but we acknowledge that the not-at-issue requirement may behave more like a post-supposition (e.g., Brasoveanu 2013, Charlow to appear), since it is possibly not evaluated at the point of composition.

(47) $[\![\text{ma}]\!] = [\lambda p.\, \lambda x\colon\ \text{PR}(x).\, [\lambda q.\, q = p]]$

The paradigmatic distinctions between QS and NQS (or between IR satisfied and IR not satisfied) in their embedding of *bu* and *ma* follow from the PR. Since *bu* and *ma* realize Persp^0, they introduce the PR when embedded, with the matrix subject being the perspectival center. In a matrix declarative (without modality or negation), the PR cannot be satisfied by an NQS due to its semantics, whereas it can be met by a QS:

(48) #John zhidao [xia yu le **ma**]$_{(\uparrow)}$. *NQS: ✗PR*
John know fall rain PRF MA
Intended: 'John knows whether it rained.'

a. John knows the answer to the question of whether it rained.

b. It is not possible that John does not know the answer.

c. The question of whether it rained is not p-active for John.

(49) John xiang zhidao [xia yu le **ma**]$_{\uparrow}$. *QS: ✓PR*
John want know fall rain PRF MA
'John wants to know whether it rained.' (Bhatt & Dayal 2020: 1122, adapted)

a. John wants to know the answer to the question of whether it rained.

b. It is possible that John does not know the answer.

c. The question of whether it rained is p-active for John.

Concretely, we propose that question embedding predicates, be they QS or NQS, can freely select a ForceP or a PerspP. There is nothing in the syntax that prevents a PerspP from being the complement of either type. Yet, an NQS is incompatible with contexts in which the centered question is p-active for the perspectival center: one cannot possibly be ignorant about the answer to a question when they already know it.

4.3 Necessity for both components

The presence of the PR demands a conspiratorial theory of selection, where syntax and semantics both contribute to the (un)acceptability of clausal complementation. Besides the effect of matrix modality, negation, and interrogative discussed in Section 3.1, another piece of evidence that semantics plays a role in interrogative complementation has to do with the fact that the way to say 'wonder' in Mandarin is lexically composed of 'want' and 'know'. Should selection be purely syntactic here, 'want to know' would share the same embedding profile as 'know'. This has shown to be not true, as we have seen in Sections 3.1 and 4.2, where QS and NQS differ in their sensitivity to the PR.

The semantic component of our proposal does not undermine the necessity for a syntactic distinction between ForceP and PerspP. An alternative account that avoids the splitting of PerspP from ForceP would predict that *bu* and *ma* questions can be embedded in any question embedding context as long as the context satisfies the PR, including noun complements and clausal subjects. This prediction is not borne out. In (50), for instance, the perspectival center — in this case, the speaker — is ignorant of the answer to the question, but the sentence is still unacceptable.

(50) *Noun complement or clausal subject*
*[Mingtian xia yu **bu/ma**]$_{(\uparrow)}$ (de wenti) hai bu qingchu.
tomorrow fall rain BU MA COMP question yet not clear
Intended: '(The question of) whether it will rain tomorrow is not clear yet.'

The same observation can be made with clausal objects of prepositions and with unconditionals. The p-activeness of the embedded question for the question holders in (51–52) does not rescue the sentence from the unacceptability of embedding *ma* questions.

(51) *Clausal object of a preposition*
Xiaohong dui [zuotian xia-le yu **meiyou**/***bu**/***ma**] hen haoqi.
Xiaohong about tomorrow fall-PFV rain MEIYOU BU MA very curious
(Intended:) 'Xiaohong is very curious about whether it will rain tomorrow.'

(52) *Unconditional*
Wulun [wo jige-le **meiyou**/***bu**/***ma**], wo dou xiwang ni gaosu wo.
no.matter 1SG pass-PFV MEIYOU BU MA 1SG all hope 2SG tell 1SG
(Intended:) 'No matter if I passed, I hope you will tell me.'

In the current account, we attribute the observations in (50–52) to the requirement on the clause size in the corresponding question embedding contexts: a PerspP is too large for these contexts to embed. Consequently, *bu* and *ma* are banned from these contexts, regardless of the epistemic status of the perspectival center.

4.4 Derivations of the question markers' (un)embeddability

With the two components established, we derive the (un)embeddability patterns of all aforementioned question markers. *Wh*-questions, *meiyou*, and A-not-A are all completely embeddable. They do not introduce the PR, and we consider them to be of the ForceP size. The derivation of their embeddability is illustrated using *meiyou* in (53–54).

(53) *Embedded* meiyou *question as noun complement or clausal subject*
[Zuotian xia-le yu **meiyou**] (de wenti) hen guanjian.
Zhangsan fall-PFV rain COMP question very crucial
'(The question of) whether it rained yesterday is very crucial.'

a. Noun complements and clausal subjects must be no larger than ForceP.
b. A *meiyou* question is a ForceP and not larger than ForceP (✓syntax).
c. (53) is acceptable.

(54) *Embedded* meiyou *question as clausal object*
Xiaohong$_i$ (xiang) zhidao [ta$_i$ jige **meiyou**].
Xiaohong want know 3SG pass MEIYOU
'Xiaohong$_i$ knows/wants to know whether she$_i$ passed.'

a. Clausal objects of question embedding predicates must be no larger than PerspP.
b. A *meiyou* question is a ForceP and not larger than PerspP (✓syntax).
c. (54) is acceptable.

As for *bu*, *ma*, and *ne* questions, they are not embeddable as noun complements or clausal subjects due to their size being PerspPs and exceeding ForcePs. When embedded as clausal objects, they introduce the PR. We illustrate their derivation using *ma* in (55–57).

(55) *Embedded* ma *question as noun complement or clausal subject*

*[Zuotian xia-le yu **ma**]$_{(\uparrow)}$ (de wenti) hen guanjian.
yesterday fall-PFV rain MA COMP question very crucial
Intended: '(The question of) whether it rained yesterday is very crucial.'

a. Noun complements and clausal subjects must be no larger than ForceP.
b. A *ma* question is a PerspP and larger than ForceP (✗syntax).
c. (55) is unacceptable.

(56) *Embedded* ma *question as clausal object of NQS (in a matrix declarative without modality or negation)*

\#Xiaohong$_i$ zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_{\uparrow}$.
Xiaohong know 3SG POSS score than 1SG high MA
Intended: 'Xiaohong$_i$ knows whether her$_i$ score is higher than mine$_j$.'

a. Clausal objects of question embedding predicates must be no larger than PerspP.
b. A *ma* question is a PerspP and not larger than PerspP (✓syntax).
c. A *ma* question introduces the PR.
d. Since the matrix subject knows the answer, the PR is not satisfied (✗semantics).
e. (56) is unacceptable.

(57) *Embedded* ma *question as clausal object of QS (in an matrix declarative without modality or negation)*

Xiaohong$_i$ xiang zhidao [ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_{\uparrow}$.
Xiaohong want know 3SG POSS score than 1SG high MA
'Xiaohong$_i$ wants to know whether her$_i$ score is higher than mine$_j$.'

a. Clausal objects of question embedding predicates must be no larger than PerspP.
b. A *ma* question is a PerspP and not larger than PerspP (✓syntax).
c. A *ma* question introduces the PR.
d. Since the matrix subject wants to know the answer, the PR is satisfied (✓semantics).
e. (57) is acceptable.

Ba questions are unembeddable across the board. We attribute their unembeddability to their exceedingly large clause size and illustrate this in (58–59).

(58) *Embedded* ba *question as noun complement or clausal subject:* ✗

*[Zuotian xia-le yu **ba**] (de wenti) hen guanjian.
yesterday fall-PFV rain BA COMP question very crucial
Intended: '(The question of) whether it rained yesterday is very crucial.'

a. Noun complements and clausal subjects must be no larger than ForceP.
b. A *ba* question is a CommitP and larger than ForceP (✗syntax).
c. (58) is unacceptable.

(59) *Embedded* ba *question as clausal object:* ✗

*Xiaohong$_i$	(xiang)	zhidao	[ta$_i$	de	fenshu	bi	wo$_j$	gao	**ba**].
Xiaohong	want	know	3SG	POSS	score	than	1SG	high	BA

Intended: 'Xiaohong$_i$ knows/wants to know whether her$_i$ score is higher than mine$_j$.'

a. Clausal objects of question embedding predicates must be no larger than PerspP.

b. A *ba* question is a CommitP and larger than PerspP (✗syntax).

c. (59) is unacceptable.

5 Different wines in different vessels

We supplement our split iForceP proposal and strengthen our semantic account of the non-co-occurrence of question markers by fully articulating our implementation of their semantics. Ultimately, we arrive at the paradigm presented in Table 6. Along the way, we present evidence that any plausible implementation of the semantics of Mandarin question markers should reflect two distinctions — namely, (i) an assertion-question distinction (as discussed in Section 4.1) and (ii) a monopolar-bipolar distinction in polar questions. The first two upcoming subsections serve the first point, and the next two serve the second.

Table 6 Inputs and outputs of question markers
Note: Q_{PL} represents a plural propositional set.

ForceP	PerspP	CommitP
$p \rightarrow$ *wh-* $\rightarrow Q_{\text{PL}}$	$p \rightarrow bu \rightarrow \{p, \neg p\}$	$p \rightarrow ba \rightarrow p$
$p \rightarrow$ *meiyou* $\rightarrow \{p, \neg p\}$	$p \rightarrow ma \rightarrow \{p\}$	
$p \rightarrow$ A-not-A $\rightarrow \{p, \neg p\}$	$Q_{\text{PL}} \rightarrow ne \rightarrow Q_{\text{PL}}$	

We begin by introducing five documented assertiveness diagnostics and extending them to Mandarin, arguing that *ba* is assertoric in nature in comparison with rising declaratives and other canonical question-forming strategies. We then use Bolinger contexts and other pieces of language-internal empirical evidence to show how the distributional distinction between *ma* questions and *meiyou*/A-not-A/*bu* questions resemble the monopolar-bipolar distinction crosslinguistically attested.

5.1 Assertiveness and nonassertiveness

We should now ask ourselves a philosophical question: What is an assertion? In response to this question, philosophers of language have provided us with a list of interrelated properties for canonical assertions, which are used to design assertiveness diagnostics.

First, a canonical assertion fully commits the speaker to its propositional content (e.g., Searle 1979, Krifka 2015). Thus, when a speaker makes two assertions that have opposite propositional contents, a contradiction arises, and the discourse is considered aberrant (60a). On the other hand, canonical questions — genuine information-seeking questions that carry no bias — do not make public what the speaker commits. In this case, even if two canonical polar questions have completely incompatible sentence radicals, they can be uttered in a sequence (60b).

(60) *Contradiction*

a. John is in Beijing. #John is in Shanghai. *canonical assertion:* ✓

b. Is John in Beijing? Is John in Shanghai? *canonical question:* ✗

Second, canonical assertions are capable of, and are typically used for, proffering information (Bach & Harnish 1979). This specific type of information proffering, which Roberts (2012, 2023) refers to as *proposition proffering*, allows certain discourse moves by the addressee that are not allowed by questions. For example, Rudin (2022) observes that English assertions but not polar questions can be replied with an expression of gratitude for the speaker's act of giving the information (e.g., "thanks for the heads up") or with an indication of receiving the previously unknown information (e.g., "oh, I had no idea"), as shown in (61–63).[20]

(61) *Context for (62–63)*
(Alvin is looking at Facebook on his phone, where he sees a cryptic post by his friend Carrie, which seems to suggest that she's been fired from her job. He turns to Bertha, who is close with Carrie, and says: …) (Rudin 2022: 344)

(62) *Canonical assertion:* ✓*Proposition proffering*
A: Carrie got fired.

a. B: Thanks for the heads up.

b. B: Oh, I had no idea.

(Rudin 2022: 344–345)

(63) *Canonical question:* ✗*Proposition proffering*
A: Did Carrie get fired?

a. B: #Thanks for the heads up.

b. B: #Oh, I had no idea.

(Rudin 2022: 344)

Third, a canonical assertion assumes that the speaker obtains knowledge about or at least belief in what they assert (Williamson 1996). As such, when an assertion is accepted, the addressee is not obliged to provide more information. A canonical question, on the other hand, does request for more information, and once accepted, it puts the addressee under the obligation to provide as much information as they can to resolve it (Roberts 2012). Rudin (2022) tests this contrast by setting up a context in which the addressee raises a related issue in response to the speaker's utterance (64). In this configuration, the addressee's acceptance of the move made by the speaker is indicated by the absence of overt rejection. When the speaker utters an assertion, raising

[20] There are cases where a question seems to be used for proffering information, too. A classic example is below:

(i) By the way, did you know that Jack Robinson was my cousin? (Bolinger 1979: 88)

Bolinger's (1979) example can apparently be responded with "oh, I had no idea," but it cannot be responded with "thanks for the heads up." Anticipating the diagnostics introduced in the upcoming Mandarin subsection, we also note that "I see" is not a felicitous response. In light of these observations, the felicity of "oh, I had no idea" may arise either from accommodation or from the recognition that it resolves the matrix question to some extent, since it entails that the responder does not know the answer. Bolinger also observes that this type of noncanonical questions typically occur with factives. In this section, we avoid using factives in our examples so as not to trigger such noncanonical interpretations.

a related issue is considered natural (65), whereas when the speaker utters a question, the same response is considered unacceptable (66). Rudin therefore concludes that questions but not assertions solicit answers.

(64) *Context for (65–66)*
(Alvin is looking at Facebook on his phone, where he sees a cryptic post by his friend Carrie, which seems to suggest that she has a new girlfriend. He turns to Bertha, who is close with Carrie, and says: …)
(Rudin 2022: 345)

(65) *Canonical assertion: ✗Answer solicitation*
A: Carrie got a new girlfriend.
a. B: Yeah, she told me about it this morning.
b. B: I don't think so. Maybe she's just trying to stir up drama.
c. B: Did you know Delia is leaving her husband?
(Rudin 2022: 345)

(66) *Canonical question: ✓Answer solicitation*
A: Did Carrie get a new girlfriend?
a. B: Yeah, she told me about it this morning.
b. B: I don't think so. Maybe she's just trying to stir up drama.
c. B: #Did you know Delia is leaving her husband?
(Rudin 2022: 345)

For the sake of completeness, we draw on two more diagnostics frequently applied to differentiate assertions and questions: the 'after all' test and the 'tell me' test (Sadock 1971, 1974, Asher & Reese 2005, 2007, Reese & Asher 2007, 2009, Kamali 2020, X. Yuan 2021). Based on the observation by Sadock (1971: 225–227) that *after all* co-occurs with declaratives much more freely than questions, Asher & Reese (2005, 2007) and Reese & Asher (2007, 2009) generalize that sentence-initial *after all* can precede assertions (67a) but not neutral questions (67b).

(67) *'After all'*
It is fine if you don't finish the article today.
a. After all, your adviser is out of the country. *canonical assertion:* ✓
b. #After all, is your adviser out of the country? *canonical question:* ✗
(Reese & Asher 2009: 157)

As for the 'tell me' test, Sadock (1974: 112) observes that true questions allow this imperative pretag (68a), whereas assertions do not (68b).

(68) *'Tell me'*
a. #Tell me, John owns a car. *canonical assertion:* ✗
b. Tell me, does John own a car? *canonical question:* ✓
(Reese & Asher 2009: 158)

Each of these diagnostics should be applied with caution, especially when not accompanied by other tests, but when considered together, their results help reveal the semantic flavor of different structures. A summary of the diagnostics covered in this subsection is presented in Table 7, where we add results of applying these diagnostics to English tag questions (cf. Asher & Reese 2007, Reese & Asher 2007, 2009) and RDs, which will be relevant to the discussion in Section 5.2.[21]

Table 7 Assertiveness diagnostics in English

	Contradiction	Proposition proffering	Answer solicitation	'After all'	'Tell me'
Canonical assertion	✓	✓	✗	✓	✗
Canonical question	✗	✗	✓	✗	✓
Tag question	✓	✗	✓	✓	✓
RD	✗	✗	✓	✗	✗

5.2 Applying assertiveness diagnostics in Mandarin

In this subsection, we apply the assertiveness diagnostics in Mandarin. While the statuses of declaratives, *wh*-questions, and nonconfirmation polar questions are relatively clear in the literature, *ba* and RDs require further investigation. As the semantic nature of these question-forming strategies is now at issue, we refrain from providing free translations for examples of *ba* and RDs in this subsection, even though we have done so in previous sections to illustrate their uses. Additionally, anticipating our conclusion, we make the following adjustments and clarifications to our presentation and terminology related to *ba*: (i) we replace the term "*ba* questions" with "*ba* utterances"; and (ii) we punctuate *ba* utterances with full stops, but we do not mean to suggest any distinction in the use of *ba* between the *ba* utterance examples in this subsection and the *ba* confirmation question examples discussed thus far.

We begin with the contradiction test, where we provide a declarative (69a) and a *ma* question (69b) as baselines. We observe that *ba* utterances (69c) behave like declaratives, whereas RDs (69d) behave like *ma* questions. (See also X. Yuan 2021: 912 and Yang 2021: 529 for other examples of this test applied to *ba*.)

(69) *Contradiction*

a. Xiaoming zai Beijing. #Xiaoming zai Shanghai. *declarative:* ✓
Xiaoming at Beijing Xiaoming at Shanghai
'Xiaoming is in Beijing. #Xiaoming is in Shanghai.'

b. Xiaoming zai Beijing **ma**? Xiaoming zai Shanghai **ma**? ma *question:* ✗
Xiaoming at Beijing MA Xiaoming at Shanghai MA
'Is Xiaoming in Beijing? Is Xiaoming in Shanghai?'

c. Xiaoming zai Beijing **ba**. #Xiaoming zai Shanghai **ba**. ba *utterance:* ✓
Xiaoming at Beijing BA Xiaoming at Shanghai BA

d. Xiaoming zai Beijing? Xiaoming zai Shanghai? *RD:* ✗
Xiaoming at Beijing Xiaoming at Shanghai

[21] In Table 7, judgments of RDs in the contradiction test are reported based on contexts where the speaker makes a random guess or receives mixed information. For more discussion on the properties of RDs, see Gunlogson 2002, 2003, 2008 and Dayal 2025.

As for proposition proffering, the paradigm we observe in (69) is fully replicated as in (70–74). Below, we use ‘thanks for letting me know’ as an expression of gratitude toward the speaker, and we use ‘I see’ as an indication of receiving previously unknown information.

(70) *Context for (71–74)*
(Angela is looking at WeChat on her phone, where she sees a cryptic post by her friend Xiaoming, which seems to suggest the he got married. She turns to Barbie, who is close with Xiaoming, and says: …)

(71) *Declarative: ✓Proposition proffering*
A: Xiaoming jiehun le.
Xiaoming marry PRF
‘Xiaoming got married.’
a. B: *Duoxie gaozhi.* ‘Thanks for letting me know.’
b. B: *Yuanlai ruci.* ‘I see.’

(72) Ma *question: ✗Proposition proffering*
A: Xiaoming jiehun le **ma**?
Xiaoming marry PRF MA
‘Did Xiaoming get married?’
a. B: #*Duoxie gaozhi.* ‘Thanks for letting me know.’
b. B: #*Yuanlai ruci.* ‘I see.’

(73) Ba *utterance: ✓Proposition proffering*
A: Xiaoming jiehun le **ba**.
Xiaoming marry PRF BA
a. B: (?)*Duoxie gaozhi.* ‘Thanks for letting me know.’
b. B: *Yuanlai ruci.* ‘I see.’

(74) *RD: ✗Proposition proffering*
A: Xiaoming jiehun le?
Xiaoming marry PRF
a. B: #*Duoxie gaozhi.* ‘Thanks for letting me know.’
b. B: #*Yuanlai ruci.* ‘I see.’

The same distinction exists in answer solicitation, but the direction of the pattern is reversed. In the same context as in (64), copied below in (75), declaratives (76) and *ba* utterances (78) do not solicit answers, but *ma* questions (77) and RDs (79) do.

(75) *Context for (76–79)*
(Alvin is looking at Facebook on his phone, where he sees a cryptic post by his friend Carrie, which seems to suggest that she has a new girlfriend. He turns to Bertha, who is close with Carrie, and says: …)
(Rudin 2022: 345)

(76) *Declarative: ✗Answer solicitation*

A: Carrie jiao-le ge xin nü pengyou.
Carrie get-PFV CLF new girl friend
'Carrie got a new girlfriend.'

a. B: *Queshi.* 'Indeed.'

b. B: *Wo tingshuo Delia ye jiao-le ge xin nü pengyou.* 'I heard that Delia also got a new girlfriend.'

(77) Ma *question: ✓Answer solicitation*

A: Carrie jiao-le ge xin nü pengyou **ma**?
Carrie get-PFV CLF new girl friend MA
'Did Carrie get a new girlfriend?'

a. B: *Queshi.* 'Indeed.'

b. B: #*Wo tingshuo Delia ye jiao-le ge xin nü pengyou.* 'I heard that Delia also got a new girlfriend.'

(78) Ba *utterance: ✗Answer solicitation*

A: Carrie jiao-le ge xin nü pengyou **ba**.
Carrie get-PFV CLF new girl friend BA

a. B: *Queshi.* 'Indeed.'

b. B: *Wo tingshuo Delia ye jiao-le ge xin nü pengyou.* 'I heard that Delia also got a new girlfriend.'

(79) *RD: ✓Answer solicitation*

A: Carrie jiao-le ge xin nü pengyou?
Carrie get-PFV CLF new girl friend

a. B: *Queshi.* 'Indeed.'

b. B: #*Wo tingshuo Delia ye jiao-le ge xin nü pengyou.* 'I heard that Delia also got a new girlfriend.'

Finally, the 'after all' (80) and 'tell me' (81) tests have been applied to Mandarin polar questions and *ba* utterances by X. Yuan (2021). We add examples of declaratives and RDs.

(80) *'After all'*

a. Bijing, ni mingtian yao qu xuexiao. *declarative: ✓*
after.all 2SG tomorrow will go school
'After all, you will go to school tomorrow.'

b. #Bijing, ni mingtian yao qu xuexiao **ma**? ma *question: ✗*
after.all 2SG tomorrow will go school MA
'After all, will you go to school tomorrow?' (X. Yuan 2021: 911, adapted)

c. Bijing, ni mingtian yao qu xuexiao **ba**. ba *utterance: ✓*
after.all 2SG tomorrow will go school BA (X. Yuan 2021: 911)

d. #Bijing, ni mingtian yao qu xuexiao? *RD: ✗*
after.all 2SG tomorrow will go school

(81) *'Tell me'*

a. #Gaosu wo, ni mingtian yao qu xuexiao. *declarative:* ✗
tell 1SG 2SG tomorrow will go school
'Tell me, you will go to school tomorrow.'

b. Gaosu wo, ni mingtian yao qu xuexiao **ma**? ma *question:* ✓
tell 1SG 2SG tomorrow will go school MA
'Tell me, will you go to school tomorrow?' (X. Yuan 2021: 911, adapted)

c. #Gaosu wo, ni mingtian yao qu xuexiao **ba**. ba *utterance:* ✗
tell 1SG 2SG tomorrow will go school BA (X. Yuan 2021: 911)

d. #Gaosu wo, ni mingtian yao qu xuexiao? *RD:* ✗
tell 1SG 2SG tomorrow will go school

We summarize (69–81) in Table 8, where we add results for *meiyou*, A-not-A, *bu*, and *ne* questions.[22] We draw three conclusions from these observations. First, *ba* utterances share their distributional pattern with canonical assertions and do not follow the pattern of canonical questions or English tag questions at all, the latter of which are frequently used to translate confirmation-seeking *ba* utterances.[23] Second, Mandarin RDs resemble canonical questions in both languages and share the same properties as English RDs. Last but not least, *ba* utterances should not be interpreted as conveying a pragmatic flavor similar to that of RDs, supporting X. Yuan (2021) and Yang (2021) but contrary to some other previous claims (e.g., Han 1995). Together, these observations support our implementation in which questions formed with *meiyou*, A-not-A, *bu*, *ma*, *ne*, or RDs are analyzed as sets of propositions, while declaratives and *ba* utterances are analyzed as propositions. From this point and onward, we refer to *ba* utterances as "*ba* declaratives."

5.3 Singleton and doubleton

We move on to establish the monopolar-bipolar distinction in Mandarin polar questions. In other languages that have been argued to exhibit this distinction, the empirical support comes from observed contrasts in (i) the (in)compatibility with unconditionals (e.g., Biezma & Rawlins 2012, Dayal 2025) and (ii) the (un)acceptability in a specific type of contexts (e.g., Krifka 2015, Matthewson 2025, Kamali 2025). The first group of empirical support is not applicable in Mandarin, given that a subset of Mandarin question markers cannot embed under unconditionals due to independent syntactic constraints, as discussed in Section 4.3. This subsection therefore mainly focuses on the second group of empirical support.

Observations about the distributional distinction between monopolar and bipolar questions date back to Bolinger (1979), who lists contexts where a plain English polar question behaves differently from its counter-

[22] Due to restrictions to be discussed in Section 5.3, *meiyou*, A-not-A, *bu*, and *ne* questions cannot be felicitously uttered in the contexts we provide for proposition proffering and answer solicitation in this subsection. In Table 8, the judgments for these question markers in these two tests are based on a controlled context in which both these questions and *ma* questions can be felicitously uttered — for example, a context where the speaker has no direct or indirect evidence and is genuinely wondering about the question they ask. A controlled context in which both questions with these question markers and declaratives/*ba* utterances/RDs can be felicitously uttered is impossible to construct.

[23] This conclusion supports a unified analysis of ba_2 (discussed in Appendix A.1) and the confirmation-seeking use of *ba* (e.g., X. Yuan 2021).

Table 8 Assertiveness diagnostics in Mandarin

	Contradiction	Proposition proffering	Answer solicitation	'After all'	'Tell me'
Declarative	✓	✓	✗	✓	✗
Meiyou question	✗	✗	✓	✗	✓
A-not-A question	✗	✗	✓	✗	✓
Bu question	✗	✗	✓	✗	✓
Ma question	✗	✗	✓	✗	✓
Ne question	✗	✗	✓	✗	✓
Ba utterance	✓	✓	✗	✓	✗
RD	✗	✗	✓	✗	✗

part in the form of an alternative question.[24] Contexts that exhibit this distinction are referred to as *Bolinger contexts*, and we follow Matthewson (2025) and abbreviate the term as *B-contexts*. An example B-context is presented in (82), where the context provides evidence against a negative response (cf. Büring & Gunlogson's (2000) *compelling contextual evidence*). Another example B-context is (83), where a positive response is the most likely, but the other alternatives are possible as well.

(82) (Somebody in your house has announced they're leaving. A little while later, you run into them in the kitchen, and you are very surprised, and you say: …) (Matthewson 2025: 30)
Are you still around (*or not)? (Bolinger 1979: 88)

(83) (Your colleague is grumpy today, and they look not great. You ask them: …) (Matthewson 2025: 33)
What's the matter? Are you tired (*or not)? (Bolinger 1979: 89)

To capture the contrast between polar questions and their alternative question counterparts, a line of work (van Rooij & Šafářová 2003, AnderBois 2011, Biezma & Rawlins 2012, Krifka 2015, 2022, 2024, Tabatowski 2022, Kamali 2025, Matthewson 2025) proposes representing the former as singletons (i.e., monopolar $\{p\}$) and the latter as doubletons (i.e., bipolar $\{p, \neg p\}$). The core insight is that a question does not have to denote all of its possible answers (departing from Hamblin 1973, for instance); rather, it denotes a set of alternative(s) that the speaker hopes to make salient (Biezma & Rawlins 2012). By asking a question, the speaker invites the answerer to choose between the salient alternatives provided by both the context and the denotation of the question. To illustrate, we begin by adopting Biezma & Rawlins's (2012) question operator, defined in (84–85).

[24] In English, an *or not* question triggers a *cornering effect* (Biezma 2009), urging the addressee to give a response, an effect not found with plain polar questions. To avoid this confound in B-contexts, in place of an *or not* question, one could instead use a *complement alternative question* (CAQ), formed by two alternatives that take up the whole possibility space (e.g., "Is the light on or off?", "Are you tired or still doing alright?", etc.). CAQs do not trigger the cornering effect (Beltrama, Meertens & Romero 2020) and still show a contrast with plain polar questions. This confound can also be avoided in languages where the cornering effect is encoded by a separate marker, and one such example is Thompson River Salish, where the cornering marker is argued to be bipolar, too (Matthewson 2025). As for Mandarin, while *ne* following a polar question has been reported to resemble English *or not* questions, as discussed in Section 3.3, we observe no other question markers examined in this paper to give rise to a cornering effect. We also discuss another SFP that has been described as producing an effect similar to cornering in Section 5.4.

(84) *Question operator*
$[\![[_{[Q]}\ \alpha]]\!]^c = [\![\alpha]\!]^c$, defined only if

a. $[\![\alpha]\!]^c \subseteq \text{SalientAlts}(c)$ or $\text{SalientAlts}(c) = \emptyset$, and
b. $|[\![\alpha]\!]^c \cup \text{SalientAlts}(c)| > 1$.

(Biezma & Rawlins 2012: 392)

(85) *Definition of SalientAlts*
$\text{SalientAlts}(c)$ is the set of propositional alternatives that are salient in the context of interpretation c.
(Biezma & Rawlins 2012: 388)

As (84b) suggests, a question can only be felicitously asked if the context and the denotation of the question together offer more than one alternative. For bipolar questions, this part of the presupposition is trivially satisfied by the two members within the alternative set represented by the question itself, but for monopolar questions, the felicity relies on the salient alternatives implicitly inferred from the context. As Matthewson (2025) puts it, a monopolar question has the alternative structure $\{p, q, r, \ldots\}$, whereas a bipolar question has the alternative structure $\{p, \neg p\}$, whose members take up the whole possibility space. This is illustrated in (86).

(86) (It has been established that there are three epistemically possible meals that the addressee may cook: pasta, stew, and fish.)

a. *Monopolar question*
Are you making pasta?

SalientAlts(c)
you are making pasta
you are making stew (implicit)
you are making fish (implicit)

b. *Bipolar question*
Are you making pasta or not?

SalientAlts(c)
you are making pasta
you are not making pasta

(Biezma & Rawlins 2012: 399–400)

Within this framework, (84a) predicts that a bipolar question cannot be felicitously asked in a context where exactly one of the two alternatives represented in its denotation is salient. The two B-contexts in (82–83) are such contexts: in each, p is salient, but $\neg p$ is not. This accounts for why *or not* questions are not felicitous in these B-contexts, while plain polar questions are.

5.4 B-contexts and more in Mandarin

Similar contrasts under B-contexts are attested in Thompson River Salish (Matthewson 2025) and Turkish (Kamali 2025), as well as in Mandarin. Krifka (2015) reports a contrast between *ma* and A-not-A questions in a B-context where a $\neg p$ response is blocked (87), a context that resembles (82).

(87) (The speaker sees the addressee eating an apple.)

a. Ma *question:* ✓
Ni chi pingguo **ma**?
2SG eat apple MA
'Do you eat apples?'

b. *A-not-A question:* ✗
#Ni **chi-bu-chi** pingguo?
you eat-not-eat apple
'Do you eat apples?'

(Krifka 2015: 11)

Krifka's observation is corroborated by Yuan & Hara's (2019) survey of sixteen native Mandarin speakers. They report that the participants have no preference between *ma* and A-not-A questions when the context accepts both p and $\neg p$ as possible answers, but they have strong preference for *ma* questions when the context blocks $\neg p$ from being a possible answer.

Prompted by Krifka and Yuan & Hara, we replicate in Mandarin the B-context contrasts observed in (82–83) below in (88–89).

(88) (Somebody in your house has announced they're leaving. A little while later, you run into them in the kitchen, and you are very surprised, and you say: …) (Matthewson 2025: 30)

a. Ma *question:* ✓
Ni hai zai zhe **ma**?
2SG still at here MA
'Are you still here?'

b. *A-not-A question:* ✗
#Ni hai **zai-bu-zai** zhe?
2SG still at-not-at here
'Are you still here?'

(89) (Your colleague looks pale today. You ask them: …)

a. Ma *question:* ✓
Ni bing-le **ma**?
2SG get.sick-PFV MA
'Did you get sick?'

b. *A-not-A question:* ✗
#Ni **bing-mei-bing**?
2SG get.sick-not.PFV-get.sick
'Did you get sick?'

With respect to the (un)acceptability under B-contexts, while RDs follow *ma* questions, *meiyou*, *bu*, and *ne* questions all pattern identically to A-not-A questions. This dichotomy lends support to the analysis that *ma* questions and RDs are monopolar, while questions formed with the other question markers are bipolar.

Aside from contrasts observed in B-contexts, we present three distributional facts in Mandarin that a monopolar-bipolar distinction naturally accounts for. These distributional facts pertain to (i) the SFP *a*, (ii) the bias marker *nandao*, and (iii) 'right' as a response.[25]

The SFP *a* (which may be realized as *na*, *nga*, *ra*, *wa*, *ya*, or *za* in different phonological contexts) conveys the speaker's attitude and emotion (Zhu 1982). Among other usages, *a* can occur in a question and convey that the speaker is requiring an answer (Chu 2009: 284–285), an effect somewhat similar to the cornering effect in English *or not* questions (Biezma 2009). As presented in (90), *a* cannot occur with *ma* or *ba*, but it can occur with *wh*-, *meiyou*, A-not-A, *bu*, and *ne* questions.

(90) a. Baozhi **shenme shihou** lai <u>a</u>?
newspaper what time come A
'When will the newspaper arrive?'

b. Baozhi lai-le **meiyou/bu/*ma/*ba** <u>a</u>?
newspaper come-PFV MEIYOU BU MA BA A
(Intended:) 'Did the newspaper arrive or not?' (Zhu 1982: 203, adapted)

[25] The adverbs *daodi* 'on earth' (Hsieh 2004, Law 2006, 2008, Her, Che & Bodomo 2022) and *jiujing* (M. Yuan 2015) have also been frequently used to illustrate a contrast between A-not-A and *ma* questions, with the claim that it can occur with the former but not the latter. However, our consultants uniformly report that they are perfectly acceptable with *ma*. This may reflect dialectal variation, but we also suggest that for speakers who allow *daodi* or *jiujing* with *ma*, it could be the case that some element in the sentence must be focused, thereby introducing a plural set of alternatives.

c. Baozhi **lai-mei-lai** (**ne**) <u>a</u>?
newspaper come-not.PFV-come NE A
'Did the newspaper arrive or not?'

To clarify, the unacceptability of *a* with *ma* and *ba* does not derive from a phonological constraint that bans two consecutive identical vowels. We can see this in (91), where *a* is allowed when following a bipolar question that ends with an identical string bearing an identical tone as *ma*.

(91) Xiaoming xiang-bu-xiang mama <u>a</u>?
Xiaoming miss-not-miss mom A
'Does Xiaoming miss his mom or not?'

The distribution of *a* can be naturally captured by positing that it semantically selects plural propositional sets. If this division holds, it aligns perfectly with the observations in English (Biezma 2009) and in Thompson River Salish (Matthewson 2025) that bipolar questions are more compatible with the cornering move, a consequence that follows from the generalization that the alternatives in the denotation of a bipolar question always exhaust the whole possibility space.

The second distributional fact concerns *nandao*, which triggers a negative bias toward the question nucleus (Xu 2017). It can occur with *ma* or RDs, but it cannot occur with *wh*-questions or other question(-like) markers (Huang, Li & Li 2009, Her, Che & Bodomo 2022), as shown in (92).

(92) a. *<u>Nandao</u> Zhangsan **weishenme** chi-le fan?
NANDAO Zhangsan why eat-PFV meal

b. <u>Nandao</u> Zhangsan chi-le fan **ma**/***meiyou**/***bu**/***ba**?
NANDAO Zhangsan eat-PFV meal MA MEIYOU BU BA
(Intended:) 'Zhangsan didn't eat, right?' (Xu 2017: 13)

c. *<u>Nandao</u> Zhangsan **chi-mei-chi** fan (**ne**)?
NANDAO Zhangsan eat-not.PFV-eat meal NE

d. <u>Nandao</u> Zhangsan chi-le fan?
NANDAO Zhangsan eat-PFV meal
'Zhangsan didn't eat, right?' (Xu 2017: 13)

This distribution can be naturally captured by positing that *nandao* semantically selects singletons.[26] Such an account aligns with Xu's (2017) view that *nandao* picks out a unique salient alternative.

[26] Apparently, *nandao* can attach either to PerspP or to CommitP, and only in the former case can it embed. This view is supported by the observation that *ma* questions with *nandao* can be quasi-subordinated, whereas RDs with *nandao* cannot:

(i) a. Xiaohong$_i$ xiang zhidao [nandao ta$_i$ de fenshu bi wo$_j$ gao **ma**]$_{\uparrow}$.
Xiaohong want know NANDAO 3SG POSS score than 1SG high MA
'Xiaohong$_i$ wants to know the answer to this question: Her$_i$ score is not higher than mine$_j$, right?'

b. *Xiaohong$_i$ xiang zhidao [nandao ta$_i$ de fenshu bi wo$_j$ gao]$_{\uparrow\uparrow}$.
Xiaohong want know NANDAO 3SG POSS score than 1SG high

Lastly, as again illustrated using a contrast between *ma* and A-not-A questions in (93–94), in Mandarin, only *ma* questions and *ba* declaratives can be felicitously answered with *dui*, *mei cuo*, or *shi de*, all of which roughly correspond to English *right* (Guo 2000, Hsieh 2001, 2004, Ye 2022, 2024).

(93) A: Ni xihuan da qiu **ma**?
2SG like play ball MA
'Do you like playing basketball?'

a. B: *Xihuan.* '[I] like [it].'

b. B: *Dui./Mei cuo./Shi de.* 'Right.'

(Ye 2024: 8)

(94) A: Ni **xi-bu-xihuan** da qiu?
2SG like-not-like play ball
'Do you like playing basketball?'

a. B: *Xihuan.* '[I] like [it].'

b. B: #*Dui.*/#*Mei cuo.*/#*Shi de.* 'Right.'

(Ye 2024: 6)

While the semantics of response particles remain under debate (Farkas & Bruce 2010, Kramer & Rawlins 2011, Krifka 2013, Holmberg 2015, Roelofsen & Farkas 2015, Farkas & Roelofsen 2019), we follow Ye (2022) and treat Mandarin response particles as propositional anaphors, which presuppose a unique alternative that is the most salient. This way, a monopolar question can license 'right' when there is no other alternative of equal salience, and it is unlikely (if not impossible) for a bipolar question to be able to license 'right' because it has two equally salient alternatives.

To recapitulate, the distributions of *nandao*, *a*, and 'right' as a response, together with the empirical findings from the B-contexts, are summarized in Table 9. Interestingly, among polar question markers, there is a strong correlation: those that contain the negative element are bipolar, whereas the only one that does not (i.e., *ma*) is monopolar. On top of the distributional observations, this correlation further motivates the semantic implementations we propose in Table 6.

Table 9 Compatibility with *a*, B-contexts, *nandao*, and 'right' as a response

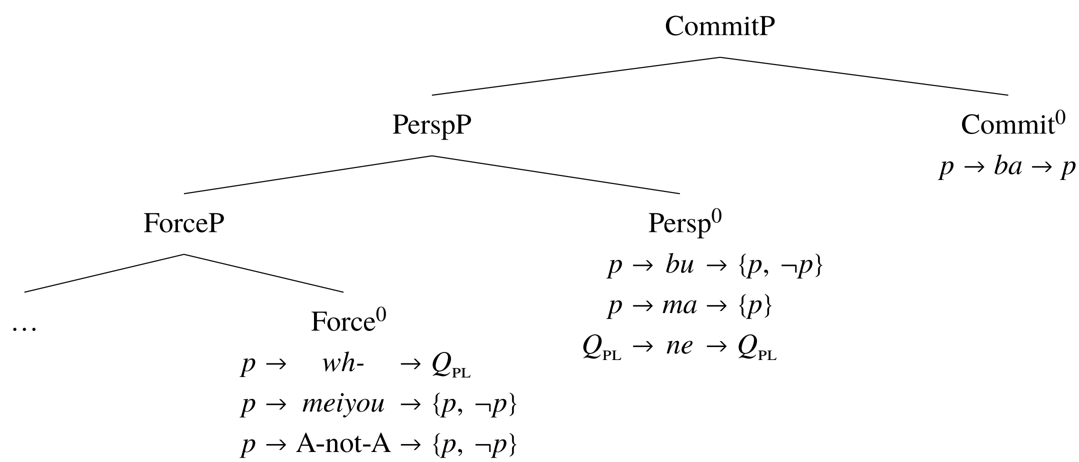
	Wh- Q_{PL}	*Meiyou* $\{p, \neg p\}$	A-not-A $\{p, \neg p\}$	*Bu* $\{p, \neg p\}$	*Ma* $\{p\}$	*Ne* Q_{PL}	*Ba* p
A	✓	✓	✓	✓	✗	✓	✗
B-contexts	✗	✗	✗	✗	✓	✗	✗
Nandao	✗	✗	✗	✗	✓	✗	✗
'Right' as a response	✗	✗	✗	✗	✓	✗	✓

6 Conclusion

The embeddability profiles of Mandarin question markers divide them into three groups: (i) the always embeddable ones; (ii) the restrictedly embeddable ones, which are sensitive to the PR; and (iii) the never embeddable ones. This trichotomy contradicts the traditional view on the syntactic statuses of these question markers that categorizes them using a root-nonroot distinction, highlighting a third possibility that warrants a conspiracy of syntax and semantics for the theory of complement selection. As such, Dayal's (2025) three-way split in the left periphery is corroborated to be systematically manifested in Mandarin. In the current implementation, ForceP marks the earliest stage of clause-typing, PerspP triggers perspectival centering, and CommitP performs commitment modifying.

The non-co-occurrence of question markers is subject to the semantic types of their input and output rather than solely affected by a syntactic competition. When scrutinized under assertiveness diagnostics, they show a contrast that is characteristic of that between canonical assertions and questions, revealing that some seemingly homogeneous question markers are in fact different in their semantic nature. Aside from an assertion-question distinction, B-contexts and the distributional facts of various question-related elements motivate a monopolar-bipolar distinction, providing novel typological evidence for a single-nonsingleton contrast in question denotations. Put together, the syntactic and semantic modules of the analysis yield a comprehensive account for the Mandarin question markers, as summarized in (95).

(95)



A Appendix: Different uses of SFPs

Debates have long centered on whether certain question markers under discussion exhibit functions beyond question marking. This appendix reviews previous claims concerning *ma*, *ne*, and *ba*.

A.1 Different uses of *ma* and *ba*

The relevant data for *ma* and *ba* are provided below, where we use subscripts to indicate distinct usages of a phonologically identical SFP without necessarily implying a theoretical commitment as to whether these represent the same or different lexical items:

(96) a. Ta bei she yao-guo **ma₂**. *declarative* + ma₂
3SG PASS snake bite-EXP MA₂
'You see, he has been bitten by a snake.' (EXP = experiential) (Pan 2019: 69)

b. Jinlai **ma₂**! *imperative* + ma₂
Come.in MA₂
'Just come in!' (Kim 2019: 19, adapted)

c. Ni zai shuo shenme **ma₂**? *interrogative* + ma₂
2SG PROG say what MA₂
'Just tell me, what are you saying?' (Kim 2019: 24, adapted)

(97) a. Ta zuotian yinggai lai-guo **ba₂**. *declarative* + ba₂
3SG yesterday should come-EXP BA₂
'Probably, she came yesterday.' (EXP = experiential) (Pan 2019: 69)

b. Ni xiang-yi-xiang **ba₃**! *imperative* + ba₃
2SG think-one-think BA₃
'Why don't you think about it a little?' (Li & Thompson 1981: 308)

c. Ni daodi yao shenme **ba₄**? *interrogative* + ba₄
2SG on.earth want what BA₄
'Just tell me, what on earth do you want?' (Chao 1965: 807)

Ma₂ "expresses that the reason is obvious" (Ding et al. 1961: 215; cf. Chappell 1991), "assumes that the other person doesn't know" (Zhu 1982: 213), or operates as "a speech act intensifier" (Kim 2019: 5), while *ba₂* "expresses a speculative tone, somewhere between doubt and belief" (Ding et al. 1961: 213), *ba₃* is used for "requesting, advising, urging, and commanding" (Ding et al. 1961: 211), and *ba₄* occurs "in impatience scenarios" (Ettinger & Malamud 2019: 245) and carries "an 'unfriendly' effect" (X. Yuan 2020: 470). Chao (1965) and Zhu (1982) treat *ba₃* and *ba₄* uniformly as an "advisative particle" (Chao 1965: 809) or "imperative marker" (Zhu 1982: 211), noting that the imperative force lies in prompting the addressee to provide an answer when *ba₃₋₄* is attached to an interrogative. Other analyses that treat *ba₂₋₄* and the confirmation question marker *ba* uniformly suggest that they as one lexical item convey "the uncertain mood" (Hu 1981b: 416, cf. Chu 1998, 2009, Chu & Li 2004), have "the effect of soliciting the approval or agreement of the hearer" (Li & Thompson 1981: 307), or "weaken the speaker's commitment (i.e., neustic)" (Han 1995: 100, cf. B. Li 2006, Kim 2019). See also X. Yuan 2021 for a unified analysis of *ba* and *ba₂* and Yang 2021 for *ba₂* and *ba₄*. Notably, while *ba* and *ba₂₋₄* are usually written with the same character, *ma₂* can be written in multiple ways, all of which use a different character from the polar question marker *ma* and potentially correspond to phonologically distinct but similar forms, such as *me* as documented by Chao (1926, 1965), Ding et al. (1961), and Chappell (1991) and *mo* as documented by M.-T. Liu (1964), Hu (1981b), and Zhu (1982).

X. Yuan (2021) further notes that the interpretation of *ba*/*ba₂* depends on whether the accompanying intonation is falling or rising. It is therefore important to clarify that all *ba* examples cited or provided in this paper concern cases with falling intonation, even when a question mark is used to highlight the confirmation-seeking use of certain examples. In the absence of experimental evidence clarifying the relationship between the intonation found in rising *ba*/*ba₂* utterances and that in rising declaratives or that typically associated with information-seeking polar questions (e.g., *ma* questions), we leave rising *ba*/*ba₂* utterances for future work.

Although detailed analyses of *ma*$_2$ and *ba*$_{2–4}$ lie beyond the scope of this paper, as discussed in Footnotes 14 and 23, our proposal adopts a stance on a part of these debates. Our arguments favor the position that *ma*$_2$ and the polar question marker *ma* are distinct lexical items, and *ba*$_2$ and the confirmation question marker *ba* are the same lexical item, while remaining agnostic about the status of *ba*$_{3–4}$.

A.2 Different uses of *ne*

Like *ma* and *ba*, *ne* wears many hats. When used sentence-finally, *ne*$_2$ "exaggerates the matter" (Ding et al. 1961: 215; cf. Lü 1980, 1982, Wu 2005, Dong 2018), expresses an "interest in additional information" (Chao 1965: 805), or "calls the other person's attention to a specific point in what the speaker is saying" (Hu 1981b: 417; cf. S.-W. Tang 2010b) in declaratives, while *ne*$_3$ conveys a continuous aspect (e.g., Chao 1965, Lü 1980, Zhu 1982). Neither causes any confound in (39–40) or the other *ne* examples in this paper, as *ne*$_2$ occurs only in declaratives (cf. Chu 2009, X. Yuan 2023 for a unified treatment of *ne* and *ne*$_2$), and *ne*$_3$ is positioned lower than the projection of the question marker *ne* in all existing cartographic studies of the Mandarin left periphery that make a distinction between them (e.g., Paul 2014, Pan 2015). Additionally, since *ne*$_3$'s meaning is incompatible with the perfective aspect, it cannot co-occur with *meiyou* or the verbal suffix *-le*.

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